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Digital Twin for railway network simulation

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Contents

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Artificial Intelligence Usage Disclaimer

Throughout the preparation of this thesis, artificial intelligence tool named ChatGPT was used as support tools. Its use is described below in full transparency.

Writing assistance ChatGPT was used during the writing process to help generate initial drafts and improve the quality of the text. For each section, I first provided the main ideas, the structure, and the technical elements that had to be included. The generated text was then reviewed, corrected, and rewritten to ensure that it accurately reflected the work carried out. It was also used to correct grammatical and orthographical errors and to improve the clarity and readability of the text. The ideas, structure, and technical content remain my own.

Literature search ChatGPT was used to help identify relevant research papers, technical documentation, and related work. However, all the sources used in this thesis were independently read, verified, and evaluated before being included.

Implementation ChatGPT was used to assist with debugging and with the implementation of repetitive or standard parts of the code. It was also used to suggest possible corrections when errors were encountered. The overall architecture of the pipeline, the main design choices, and the algorithms developed in this work were designed and implemented independently.

Limitation of use ChatGPT was not used to define the research question, determine the research methodology, interpret the experimental results, or draw the final conclusions of this thesis. It was used as a support tool throughout this work and not as a substitute for independent thinking, analysis, or scientific judgment.

Chapter 1

Introduction

Railway networks are essential infrastructures for passenger mobility, freight transport, and sustainable transportation. In dense networks such as the Belgian railway system, operating trains efficiently and reliably is a complex task. Many trains, stations, tracks, and operational constraints interact continuously, while disruptions in one part of the network can propagate to other services and affect punctuality at a wider scale.

This complexity is both infrastructural and operational. Trains must follow predefined routes and timetables, share limited track capacity, respect safety constraints, and interact with elements such as switches, platforms, and stations. As a result, railway operators and researchers increasingly rely on digital tools to better understand, simulate, and analyse railway systems.

In this context, the concept of a digital twin has gained increasing attention. Informally, a digital twin can be understood as a virtual counterpart of a physical system, connected to data and used to observe, simulate, and analyse its behaviour. Applied to railways, such a representation may combine infrastructure data, train movements, operational constraints, and traffic conditions. Simulation is therefore a central component of this vision, since it provides a controlled environment in which railway operations can be reproduced, tested, and compared with real data.

This thesis focuses on the construction of a simulation-oriented railway digital twin using SUMO. The objective is not to build a complete real-time digital twin, but to study how heterogeneous railway infrastructure and operational datasets can be transformed into simulation-ready artefacts. The work also investigates how different levels of railway network representation influence the feasibility and realism of the resulting simulation.

1.1 Research Problem and Objectives

Despite the availability of railway data and simulation platforms, constructing a realistic railway simulation from real-world datasets remains difficult. Railway data is heterogeneous by nature: infrastructure datasets may describe stations, tracks, switches, platforms, and geographical geometries, while operational datasets describe train movements, schedules, arrival and departure times, and punctuality records. These datasets do not always share the same structure, granularity, or level of detail. Integrating them into a single simulation framework therefore requires several processing, modelling, and approximation steps.

A first challenge concerns the representation of the railway infrastructure. At a high level, the network can be represented as a graph in which stations are nodes and railway links are edges. This abstraction is useful because it simplifies route reconstruction and supports large-scale simulation. However, it does not represent detailed infrastructure elements such as platforms, switches, parallel tracks, or station-level routing constraints.

A second challenge appears when moving toward a more detailed representation. At the micro level, tracks, platforms, switches, and routing structures must be represented explicitly. This provides a more realistic description of the infrastructure, but it also requires a topologically consistent network. Platforms must be correctly associated with tracks, and valid paths must exist between consecutive stations. Even small inconsistencies in the reconstructed infrastructure can prevent trains from being routed correctly.

A third challenge concerns the integration of real operational data. Punctuality data provides information about train stops, planned and observed times, and delays, but it does not directly provide the exact physical route followed by each train. Train journeys must therefore be reconstructed from station sequences and mapped onto the generated simulation network. This process is relatively robust at the macro level, but becomes more fragile at the micro level because each movement must correspond to a valid sequence of physical track segments.

These challenges are addressed through the following research sub-questions:

RQ1. How can heterogeneous railway infrastructure and operational datasets be transformed into SUMO-compatible simulation components?

RQ2. To what extent can a macro-level station-to-station representation support the integration of real operational data into railway simulation?

RQ3. To what extent can a micro-level infrastructure representation support detailed railway simulation based on tracks, platforms, switches, and routing structures?

RQ4. What are the main limitations and failure points encountered when integrating real operational data into micro-level railway simulation?

The main objective of this thesis is to design and implement a data-driven pipeline for constructing a simulation-oriented railway digital twin. This pipeline is evaluated through two complementary modelling approaches. The macro-level model represents the railway network as a station-to-station graph and is used to validate the integration of real operational data into simulation. The micro-level model represents the infrastructure in greater detail and is used to study the feasibility of more realistic railway simulation.

The expected outcome is therefore not a perfect reproduction of the Belgian railway network, but a structured and evaluated prototype showing how railway data can be transformed into SUMO-compatible simulation models. The contribution lies in the construction of the modelling pipeline, the comparison between macro-level and micro-level representations, and the identification of the technical challenges that must be addressed to progress toward more complete railway digital twins.

1.2 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 presents the state of the art. It introduces the concept of digital twins, first from a general perspective and then in the context of railway systems. It also reviews the main components of railway infrastructure and operations, discusses common approaches for railway network representation, and presents existing railway simulation tools, with particular attention to SUMO.

Chapter 3 describes the proposed methodology. It positions the work with respect to the research gaps identified in the literature, presents the overall framework of the approach, justifies the choice of SUMO, and describes the infrastructure and operational datasets used in the implementation.

Chapter 4 details the implementation of the proposed approach. It is divided into two main parts. The first presents the macro-level model, including data processing, network generation, trip generation, and simulation. The second presents the micro-level model, including infrastructure modelling, switch detection, platform detection, track segmentation, dual graph construction, trip generation, and simulation.

Chapter 5 presents the results obtained from the implemented models. It analyses dataset coverage, infrastructure modelling quality, simulation results, and computational performance. Particular attention is given to the difference between successful macro-level integration and the difficulties encountered when using real operational data at the micro level.

Chapter 6 discusses the main limitations of the proposed methodology. It provides a more detailed analysis of issues related to data quality, modelling assumptions, route reconstruction, SUMO constraints, and scalability.

Chapter 7 concludes the thesis by summarizing the main contributions and findings. It also discusses the practical implications of the work and presents future research directions, including counterfactual simulation, track work planning, real-time forecasting, incident propagation, and framework generalization.

Chapter 2

State of the Art

2.1 Digital Twins

Digital twins have become a central concept in the digitalization of complex industrial systems [19]. The term is used in many domains, including manufacturing, aerospace, healthcare, smart cities, energy systems, and transportation [19, 35, 33]. Although definitions vary depending on the application domain, the general principle remains the same: a digital twin is a digital representation of a physical system connected to data and used to analyse, simulate, monitor, or support decisions about the real system [24, 19].

In the context of this thesis, the concept is particularly relevant because railway networks are complex, dynamic, and data-rich systems. They combine physical infrastructure, moving vehicles, operational rules, timetables, delays, maintenance constraints, and safety requirements. A digital twin can therefore provide a structured framework to connect infrastructure data, operational data, simulation models, and analytical tools into a coherent representation of the railway system.

2.1.1 Definition, Capabilities and Limitations

The concept of digital twin was first introduced in the context of product lifecycle management and later extended to many engineering and industrial domains. A common definition describes a digital twin as a virtual representation of a physical object, process, or system that is connected to its physical counterpart through data exchange and can be used throughout the lifecycle of the system. This definition emphasizes that a digital twin is not only a static model or a visualization tool. Its distinctive feature is the relationship between the physical system and its digital counterpart.

A digital twin can be understood through three main components: the physical system, the virtual system, and the connection between them. The physical system corresponds to the real-world object or process being represented. The virtual system describes its structure, behaviour, and state. The connection corresponds to the data flow linking both entities. This connection may be unidirectional, when the physical system only updates the virtual model, or bidirectional, when the digital model can also support decisions or actions applied to the physical system [34].

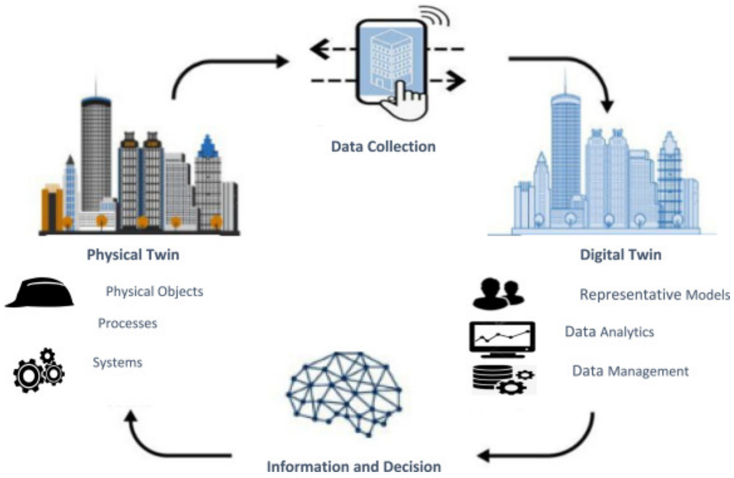

Figure 2.1: General architecture of a digital twin.

A more formal representation defines a digital twin as a system composed of several interacting elements [41, 19]:

$$DT = (P, V, D, C, S) \tag{2.1}$$

where P denotes the physical system, V the virtual representation, D the data associated with the system, C the connection mechanisms between the physical and virtual entities, and S the services built on top of the twin, such as simulation, prediction, visualization, diagnosis, or optimization. This representation shows that a digital twin is not only a model, but an architecture combining data, models, and services to support decision-making.

It is also important to distinguish digital twins from related concepts. A digital model is a digital representation of a system without automatic data exchange with the physical system. A digital shadow receives data from the physical system, but the information flow remains mainly one-way. A digital twin represents a more advanced form, where the digital representation is regularly or continuously updated from the physical system and can support feedback, analysis, or operational decisions [30]. In this thesis, the proposed work is closer to a simulation-oriented digital twin foundation: it focuses on constructing coherent virtual railway representations from real infrastructure and operational data, which is a necessary step before a fully real-time and bidirectional railway digital twin can be achieved.

Digital twins are usually associated with several capabilities. Monitoring consists of observing the state of a system using sensor data, logs, operational records, or information systems. Simulation uses the digital representation to reproduce the behaviour of the physical system under controlled conditions [19, 39]. Prediction uses historical or real-time data to estimate future states, failures, delays, or performance indicators [19, 35, 33]. Optimization evaluates alternative scenarios in order to support better decisions. More advanced digital twins may also include control or actuation, where decisions derived from the digital model influence the physical system.

These capabilities make digital twins attractive for complex systems, but they also introduce limitations. A digital twin is only as reliable as the data, models, assumptions, and synchronization mechanisms on which it is based. Incomplete, outdated, noisy, or inconsistent data may produce misleading results. Similarly, a model that is too simplified may ignore important physical or operational constraints, while a model that is too detailed may become computationally expensive and difficult to maintain. Validation is also a major challenge, since a digital twin must be compared with real observations, even though real systems often include hidden variables, incomplete measurements, operational exceptions, and human decisions that are not fully captured in the available data. For this reason, digital twins are often developed incrementally, starting from static representation and simulation before integrating real-time data, predictive models, and optimization services.

2.1.2 Recent Developments and Applications

The development of digital twins has accelerated with the progress of several enabling technologies. The IoTs allows physical systems to generate continuous streams of data [19]. Cloud computing and distributed architectures provide the infrastructure required to store and process large volumes of information. Artificial intelligence and machine learning make it possible to extract patterns, detect anomalies, and predict future states. Simulation tools provide a controlled environment in which alternative scenarios can be tested without disrupting the real system. Finally, visualization technologies, such as dashboards, 3D interfaces, and geographic information systems, help users interpret the state and behaviour of the DT.

Digital twins were first widely adopted in manufacturing, where they support product design, production monitoring, maintenance planning, and process optimization. In smart manufacturing, a digital twin can represent a machine, a production line, or an entire factory. It can be used to monitor equipment status, simulate production scenarios, predict failures, and improve resource allocation [39]. This domain is important in the literature because it provides many conceptual foundations later reused in other fields.

In aerospace, mechanical engineering, and energy systems, digital twins are often used to monitor critical assets and support predictive maintenance. Aircraft, engines, turbines, power plants, and renewable energy infrastructure generate large amounts of operational data during their lifecycle. A digital twin can combine this data with physics-based or data-driven models in order to estimate wear, detect abnormal behaviour, and plan maintenance interventions before failures occur [33]. Digital twins are also increasingly studied in healthcare and smart cities, where they can represent medical devices, hospital workflows, patient-specific models, urban infrastructure, mobility systems, and environmental condition [35].

These capabilities make digital twins attractive for complex systems, but they also introduce limitations. A digital twin is only as reliable as the data, models, assumptions, and synchronization mechanisms on which it is based. Incomplete, outdated, noisy, or inconsistent data may produce misleading results. Similarly, a model that is too simplified may ignore important physical or operational constraints, while a model that is too detailed may become computationally expensive and difficult to maintain. Validation is also a major challenge, since a digital twin must be compared with real observations, even though real systems often include hidden variables, incomplete measurements, operational exceptions, and human decisions that are not fully captured in the available data. For this reason, digital twins are often developed incrementally, starting from static representation and simulation before integrating real-time data, predictive models, and optimization services.

2.1.3 Applications in Railway networks

Railway systems are particularly suitable candidates for digital twin approaches. They combine physical infrastructure, operational processes, moving assets, strict safety constraints, and large volumes of data. A railway network includes tracks, switches, signals, stations, platforms, rolling stock, timetables, passenger flows, maintenance operations, and traffic management rules. These elements interact continuously, and a local disturbance can propagate through the network. Railway operators therefore require tools able to represent the state of the network, simulate traffic behaviour, evaluate disruptions, and support planning decisions [44, 44].

Railway digital twins can be applied at different levels of granularity. At the asset level, a digital twin can represent individual components such as tracks, switches, bridges, tunnels, trains, or signalling equipment. This type of twin is mainly useful for condition monitoring and predictive maintenance, where sensor data, inspection records, and maintenance logs are used to estimate asset health and anticipate interventions. At the infrastructure level, a digital twin can represent the topology of the railway network, including tracks, stations, platforms, junctions, and operational segments. This representation is necessary to analyse accessibility, capacity, routing possibilities, and infrastructure conflicts. At the operational level, a digital twin can represent train movements, schedules, delays, and traffic interactions, making it possible to analyse circulation patterns, delay propagation, and the impact of infrastructure constraints on service quality.

Several railway applications can be identified. Predictive maintenance is one of the most common, since railway infrastructure is expensive to inspect and failures can strongly affect safety and reliability. Traffic management is another important application: by reproducing train movements in a virtual environment, a digital twin can help evaluate congestion, detect bottlenecks, and test operational strategies. DTs can also support delay analysis and propagation modelling, since a delay affecting one train may influence many others. Finally, simulation-based digital twins are useful for scenario testing, such as timetable changes, temporary track closures, infrastructure works, disruptions, or emergency response strategies.

However, railway digital twins also raise specific challenges. Railway systems are safety-critical, which means that simplified models must be used carefully. Accurate representation of tracks, switches, signals, platforms, and routing constraints becomes necessary when the objective is to reproduce detailed operational behaviour. Data availability is another major issue, since infrastructure data may be incomplete, private, heterogeneous, or difficult to align with operational data. In addition, railway systems operate at several spatial and temporal scales: long-distance network connectivity, station-level infrastructure, and dynamic train operations may all need to be represented within the same framework.

For these reasons, railway digital twins often require a layered approach. A macro-level representation can model station-to-station connectivity and global train movements, while a micro-level representation can model detailed infrastructure such as tracks, switches, and platforms. This distinction is directly reflected in the methodology of this thesis. The macro-level model provides a simplified and scalable representation of the railway network, whereas the micro-level model aims to capture more detailed infrastructure constraints. Together, these two levels contribute to the construction of a simulation-oriented foundation for a railway digital twin.

In summary, digital twins provide a relevant conceptual and technical framework for railway simulation. They offer a way to combine infrastructure data, operational data, simulation models, and analytical tools into a unified system. However, building a complete railway digital twin requires several intermediate steps: reconstructing the infrastructure, generating valid train routes, integrating real operational data, validating simulation outputs, and identifying the limitations of the model. This thesis focuses on these foundational steps, with the objective of preparing the ground for future applications such as real-time forecasting, disruption analysis, and decision support.

2.2 Railway Systems

Railway systems are complex socio-technical systems composed of physical infrastructure, rolling stock, operational rules, control systems, and data-driven decision processes. Unlike road traffic, where vehicles usually have a high degree of freedom in their trajectories, trains are constrained to follow predefined tracks, respect signalling rules, and operate according to planned timetables. This makes railway systems highly structured, but also highly constrained.

Understanding railway systems is essential for the construction of a railway digital twin. A digital twin does not only require a geometric or topological description of the network. It must also represent how trains move, how infrastructure is used, how conflicts are avoided, and how delays may propagate. For this reason, the following sections introduce the main infrastructure components, operational principles, network representations, and data sources required for railway modelling and simulation.

2.2.1 Railway infrastructure

Railway infrastructure refers to the physical and technical components that allow trains to circulate safely and efficiently. It includes tracks, switches, junctions, stations, platforms, signalling systems, electrification equipment, and communication systems. From a modelling perspective, infrastructure defines both the spatial structure of the network and the constraints under which train movements can occur.

The level of detail used to represent infrastructure depends on the objective of the model. A high-level model may only represent stations and the connections between them, which is sufficient for analysing large-scale connectivity or approximate train movements. A more detailed model must represent tracks, switches, platforms, signals, and route conflicts, which is necessary for studying station-level operations, capacity, and realistic routing constraints [3].

Tracks

Tracks are the fundamental physical components of a railway network. They guide train movements and define the possible spatial paths through the infrastructure. From an operational point of view, a track is not only a physical object but also a capacity resource, since only a limited number of trains can use the same section within a given time interval [11].

In modelling terms, tracks can be represented at different levels of abstraction. At a macroscopic level, a track may be abstracted as a connection between two stations. This is useful for station-to-station connectivity and large-scale train movement analysis. At a microscopic level, tracks must be represented as individual segments with geometry, direction, length, speed constraints, and connections to neighbouring elements. This level

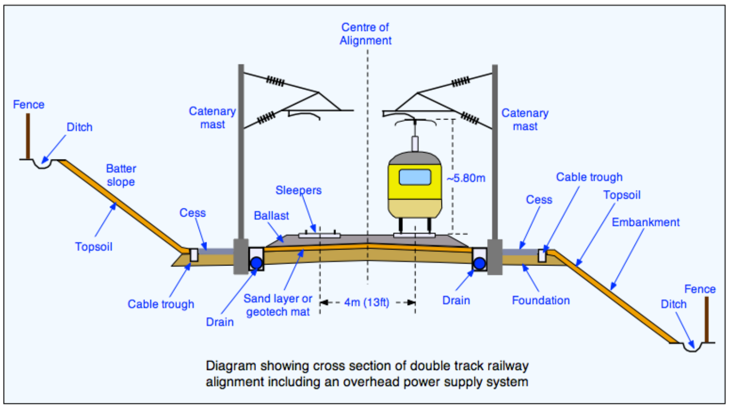

Figure 2.2: Schematic representation of railway infrastructure components.

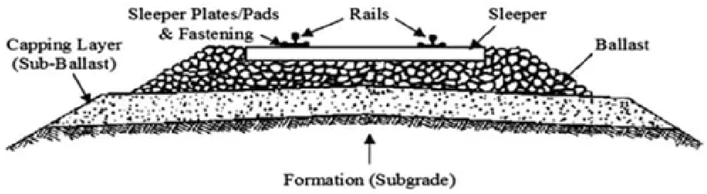

Figure 2.3: Railway track structure.

of detail is required when the objective is to compute feasible routes through complex station areas or to represent access to switches and platforms.

Track directionality is also important. Some railway lines are designed for traffic in one direction only, while others allow bidirectional operation. Even when tracks are physically bidirectional, signalling and operational rules may restrict their use. Therefore, a railway model must distinguish between physical connectivity and operationally usable connectivity.

Switches and Junctions

Switches, also called points or turnouts, allow trains to move from one track to another. They are essential for routing because they introduce branching possibilities in the network. Switches enable trains to enter stations, change tracks, access sidings, overtake other trains, or follow alternative routes [9, 10].

From a modelling perspective, a switch can be seen as a local decision point in the infrastructure graph. It connects an incoming track to several possible outgoing tracks depending on its configuration. Junctions are larger structures composed of several switches and crossings, often located near stations, yards, or intersections between railway lines.

Switches introduce important operational constraints. A train can only pass through a switch if it is correctly aligned for the intended route, and two trains cannot use conflicting routes through the same junction at the same time. These constraints are normally enforced by signalling and interlocking systems. In detailed railway simulations, switches and junctions must therefore be modelled not only as graph connections, but also as shared resources that may generate conflicts.

The correct identification of switches is particularly important for digital twin con-

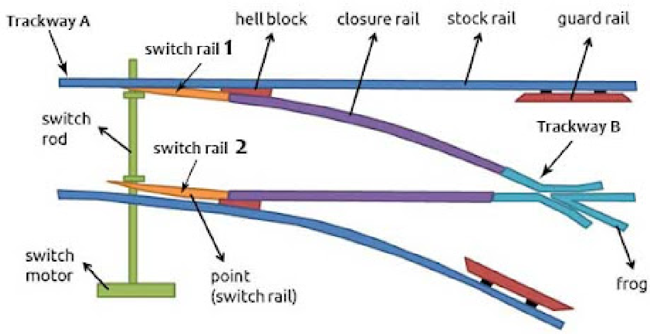

Figure 2.4: Railway switch structure.

struction. Missing switches may prevent valid routes from being found, while incorrectly detected switches may create unrealistic routing possibilities. This is one of the reasons why micro-level railway modelling is more difficult than macro-level modelling.

Stations and Platforms

Stations are operational nodes where trains may stop, passengers may board or alight, and services may be regulated. They are often among the most complex elements of a railway network because they combine tracks, platforms, switches, signals, passenger flows, and timetable constraints [20].

In a macroscopic representation, a station can be modelled as a single node. This is useful for analysing station-to-station connectivity or reconstructing train journeys from operational data. However, such a representation hides the internal structure of the station. It does not indicate which platform a train uses, whether two trains can stop simultaneously, or whether access routes to different platforms conflict.

In a microscopic representation, a station must be decomposed into tracks, platforms, stopping positions, and access routes. Platforms define where passenger exchanges occur and are important for simulation because train stops must be attached to specific tracks or lanes. If a platform is incorrectly associated with a track, the train may stop at the wrong location or may not be able to reach the stop at all.

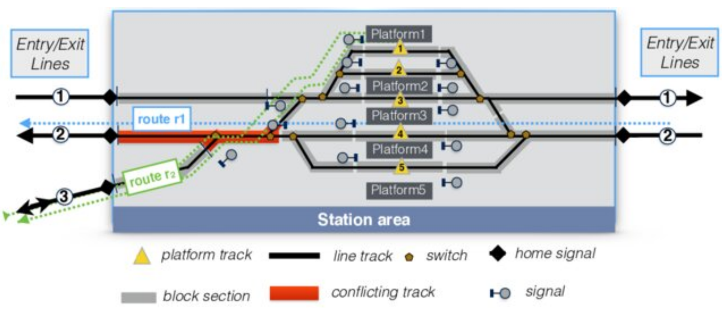

Figure 2.5: Example of a station layout

Stations also influence capacity. Dwell time, platform availability, route conflicts at station throats, and the number of available tracks determine how many trains can be handled within a given period. Large stations may become bottlenecks even if the lines around them have sufficient capacity. For this reason, station modelling is a key component of realistic railway simulation.

In this thesis, stations are used differently in the macro-level and micro-level approaches. In the macro-level model, stations are represented as nodes connected by edges. In the micro-level model, stations are associated with platforms and track segments. This distinction makes it possible to compare a simplified connectivity-based representation with a more detailed infrastructure-based representation.

Stations also influence capacity. Dwell time, platform availability, route conflicts at station throats, and the number of available tracks determine how many trains can be handled within a given period. Large stations may therefore become bottlenecks even when the surrounding lines have sufficient capacity. In this thesis, stations are represented differently depending on the modelling level: as single nodes in the macro-level model, and as platform-track associations in the micro-level model.

Signaling Systems

Signalling systems ensure the safe movement of trains through the railway network. Their role is to prevent conflicting train movements and maintain safe separation between trains. They communicate movement authorities, speed restrictions, route availability, and stopping instructions to train drivers or onboard systems [4].

Traditional signalling systems divide the railway line into block sections. Under normal operation, only one train is allowed to occupy a block section at a time. A train may proceed only if the section ahead is clear and if the route has been correctly set. This principle prevents rear-end collisions and conflicting movements. In more advanced systems, movement authority may be transmitted directly to the train through cab signalling or automatic train protection systems.

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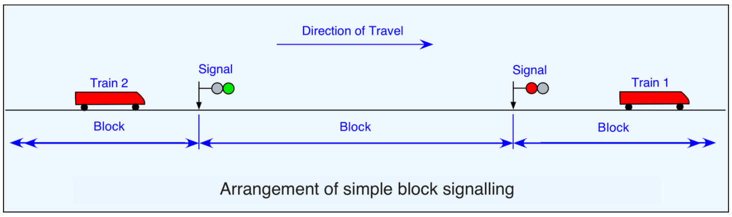

Figure 2.6: Simplified block signalling principle

In Europe, signalling and train control are strongly influenced by interoperability requirements [17]. The European Rail Traffic Management System (ERTMS) and the European Train Control System (ETCS) aim to standardize train control and signalling across countries. Such systems are relevant for digital twin research because they generate and rely on structured operational data and because they influence how train movements can be represented.

From a simulation perspective, signalling can be represented at different levels of detail. A simple simulation may ignore signalling and only check whether routes exist. A more realistic simulation must represent block occupation, signal states, route reservations, interlocking constraints, and train separation. The level of signalling detail has a direct impact on the realism of simulated train movements and capacity analysis.

In this thesis, signalling is not fully modelled as an operational control system. However, it remains an important limitation and a key concept for interpreting the realism of the generated simulations. The absence or simplification of signalling means that the simulation can validate connectivity and route generation, but cannot fully reproduce all safety-critical operational constraints of a real railway network.

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2.2.2 Railway Operations

Railway operations describe how trains are planned, dispatched, controlled, and managed over the infrastructure. While infrastructure defines what is physically possible, operations define what is actually allowed and scheduled. Railway operations combine timetables, train paths, rolling stock constraints, crew constraints, safety rules, capacity management, and real-time traffic regulation [38].

For a railway DT, operations are essential because the objective is not only to represent the network, but also to reproduce or analyse train movements. Operational data provides information about scheduled and actual train movements, delays, stops, and service patterns. This data can be used to generate simulation scenarios, validate the model, and study the behaviour of the railway system.

Train Movements

A train movement is the circulation of a train from one point of the network to another. It is defined by an origin, a destination, a route, a departure time, intermediate stops, and operational constraints. A train cannot freely choose its path at any time; its movement must be compatible with the infrastructure, timetable, signalling rules, and other train movements.

In simulation, train movements must be converted into routes and departure times. At the macro level, a route may be represented as a sequence of stations. At the micro level, it must be converted into a sequence of track segments or edges. This conversion is one of the central challenges of railway digital twin construction, because real operational records often describe train movements at station level, while the simulator requires paths through the generated network.

The feasibility of a train movement depends on both connectivity and timing. Connectivity means that a valid path must exist between consecutive locations. Timing means that the train must be inserted at the correct time and must respect operational constraints such as dwell times and travel times. If either the connectivity or timing is inconsistent, the simulation may fail or produce unrealistic behaviour.

Timetables

A railway timetable defines the planned movements of trains over the network. It specifies departure and arrival times, stopping patterns, running times, dwell times, and sometimes platform assignments or path allocations. Timetabling is a constrained optimization problem because it must satisfy service objectives while respecting infrastructure capacity, minimum headways, station capacity, rolling stock availability, and safety margins [26].

Periodic timetables are common in passenger railway systems. In such timetables, services repeat at regular intervals, for example every 30 or 60 minutes. This makes the service easier to understand for passengers and easier to coordinate across lines. However, periodicity also introduces constraints because connections, crossings, and resource usage must remain consistent across repeated time cycles.

A common way to visualize timetables is the time-space diagram [42]. In such a diagram, one axis represents time and the other represents position along a railway line. Each train is represented by a line. The slope reflects speed, stops appear as horizontal segments, and crossings or overtakes can be analysed visually. Time-space diagrams are useful for understanding capacity, conflicts, delays, and recovery margins.

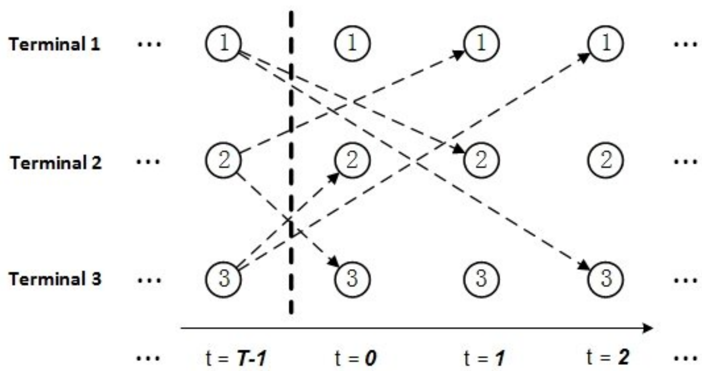

Figure 2.7: Example of a time-space diagram.

Capacity Management

Railway capacity refers to the ability of infrastructure to accommodate train movements within a given period while respecting safety, operational, and quality constraints. Unlike road capacity, railway capacity depends strongly on timetable structure, train heterogeneity, signalling, station dwell times, and route conflicts. It is therefore not only a property of the physical infrastructure, but also of how this infrastructure is used [26, 32, 23].

A railway line with homogeneous trains running at similar speeds and stopping patterns can usually accommodate more trains than a line where fast, slow, passenger, and freight trains are mixed. This is because heterogeneous traffic increases the need for headways, overtaking, and timetable margins. Similarly, a station with limited platforms or conflicting access routes may reduce capacity even if the surrounding tracks are not saturated.

The UIC 406 method is often used as a reference approach for railway capacity analysis. It evaluates capacity consumption by compressing train paths in a timetable and analysing how much time is occupied by train movements. This approach highlights that capacity is related not only to the number of trains, but also to their ordering, spacing, speed differences, and operational margins[27].

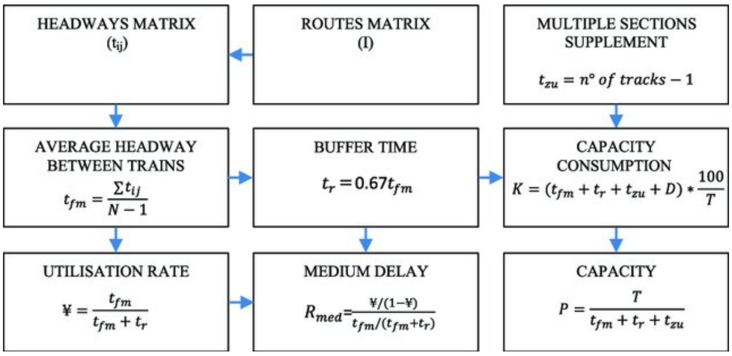

Figure 2.8: UIC 406 method procedure.

Capacity management is important for simulation because a generated route may be technically connected but operationally unrealistic if capacity constraints are ignored. For example, several trains may be assigned to the same platform or track at overlapping times. A detailed simulator should therefore account for infrastructure occupation, route conflicts, and safe separation. In this thesis, capacity is not the main optimization objective, but it remains relevant for interpreting the limitations of the macro-level and micro-level simulations.

Delay Propagation

Delay propagation is one of the most important phenomena in railway operations. A primary delay occurs when a train is delayed by an external or local cause, such as technical failure, passenger boarding time, weather, infrastructure problems, or operational incidents. A secondary delay occurs when this initial delay affects other trains through shared infrastructure, rolling stock circulation, crew dependencies, passenger connections, or timetable conflicts[21, 13].

Railway networks are particularly sensitive to delay propagation because many trains share the same tracks, stations, junctions, and timetable margins. A delayed train may occupy a track section later than planned, preventing another train from using it. It may also miss a planned crossing, delay a connection, or arrive late at a station where the rolling stock is needed for another service. As a result, a local disturbance can spread through the network.

Delay propagation can be studied at different levels. Microscopic approaches simulate train movements and conflicts in detail. Macroscopic approaches analyse delays at the level of stations, lines, or aggregated regions. Data-driven approaches use historical operational data to estimate how delays evolve over time and space. In this thesis, delay propagation is not fully modelled as a predictive process. However, reconstructing infrastructure and train movements is a prerequisite for such applications.

2.2.3 Railway Network Representation

A railway network can be represented in several ways depending on the modelling objective. The choice of representation determines which phenomena can be analysed and which assumptions are made. For digital twin and simulation purposes, three broad families of representations are particularly relevant: graph-based modelling, physics-based modelling, and logic-based modelling. These representations are not mutually exclusive, and detailed railway simulators may combine them.

Graph-Based Modeling

In graph-based modelling, the railway network is represented as a graph in which nodes correspond to stations, switches, junctions, or operational points, while edges correspond to track segments or connections between them [22]. This representation is compatible with algorithms from graph theory and operations research. It is particularly useful for routing, timetable design, resource allocation, connectivity analysis, and delay propagation studies.

The main limitation of graph-based models is that they often simplify train dynamics and operational constraints. If the graph is too abstract, trains may be treated as moving along edges without detailed acceleration, braking, signalling, or conflict constraints. Nevertheless, graph-based modelling is central to this thesis. The macro-level model uses a station-to-station graph, while the micro-level model builds a more detailed graph based on track segments and routing possibilities.

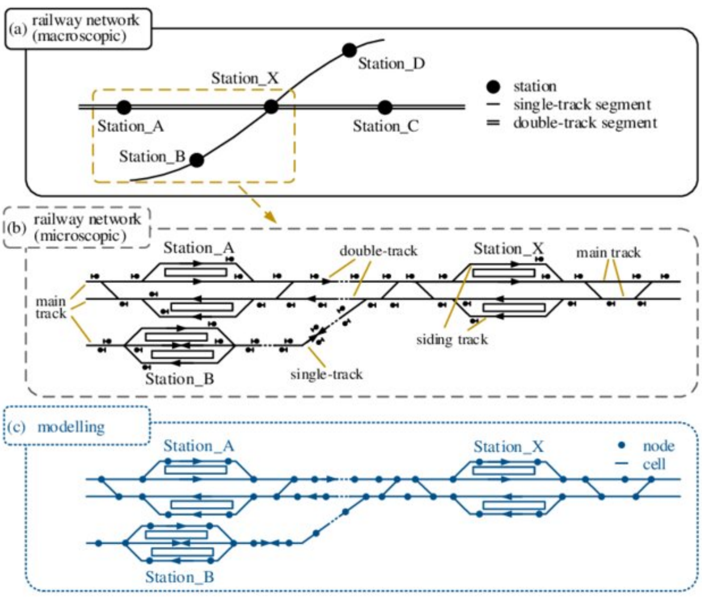

Figure 2.9: Example of a railway station modelling at macro and micro levels.

Physics-Based Modeling

Physics-based models represent the physical behaviour of trains and their interaction with the infrastructure. They simulate train motion as a function of traction force, resistance, braking, gradients, inertia, and sometimes energy consumption [18]. This approach is useful when the objective is to study train dynamics, energy-efficient driving, rolling stock performance, or precise running times.

The strength of physics-based modelling is its realism, but it requires detailed data such as train mass, traction characteristics, braking curves, gradients, and speed restrictions. It can also become computationally expensive at network scale.

Logic-Based Modeling

Logic-based models capture the discrete control rules and safety constraints of railway operations. A typical example is the modelling of interlocking systems, which ensure that only safe and conflict-free train movements are allowed. These models represent the railway as a set of states and transitions, with rules depending on the position of switches, signal states, block occupation, and route reservations.

Logic-based models are useful for validating signalling and safety-critical behaviour. They can be expressed using formal methods and model-checking tools such as UPPAAL or NuSMV [5, 2]. However, they require highly detailed rule definitions and may face scalability challenges when the network grows or when many trains operate simultaneously.

Each of the three modeling captures different but complementary aspects of a railway system. Combining graph abstraction for routing, physics modules for motion and simulation, and formal logic for safety are increasingly explored in the industry to connect these domains and support robust digital representations of railway networks.

Each modelling family captures a different aspect of the railway system. Graph-based models emphasize connectivity and routing, physics-based models emphasize motion and train dynamics, and logic-based models emphasize safety and operational rules. The work developed in this thesis mainly relies on graph-based representations, while acknowledging that more complete railway digital twins would require stronger integration of physics-based and logic-based constraints.

2.2.4 Railway Data Sources

Railway modelling and simulation depend strongly on the availability, completeness, and consistency of data. A digital twin requires data describing both the physical infrastructure and the operation of trains. These two categories are often stored in different systems, they can follow different formats, and use different levels of granularity. Aligning them is one of the main challenges of railway DT construction.

Infrastructure data describes what exists in the network, while operational data describes how the network is used. A simulation-oriented digital twin must combine both: infrastructure data is required to build the network, and operational data is required to generate train movements and validate the behaviour of the simulation.

Infrastructure Data

Infrastructure data includes information about tracks, switches, stations, platforms, signals, speed limits, electrification, geometry, and topology. Depending on the source, this data may be available as geospatial data, tabular data, XML files, engineering databases, or standardized railway formats.

OpenStreetMap and OpenRailwayMap are examples of open geospatial sources that can provide railway infrastructure information. They may include track geometry, station locations, railway line types, electrification, gauge, signals, and other attributes. These sources are useful because they are publicly accessible and cover large areas. However, their completeness and accuracy may vary depending on the region and the contributions of the mapping community.

OpenStreetMap and OpenRailwayMap are examples of open geospatial sources that can provide railway infrastructure information. They may include track geometry, station locations, railway line types, electrification, gauge, signals, and other attributes. These sources are useful because they are publicly accessible and cover large areas [7]. However, their completeness and accuracy may vary depending on the region and the contributions of the mapping community.

Institutional data sources, such as infrastructure manager databases, are usually more accurate and complete but are often private or restricted. In Belgium, infrastructure data from Infrabel or related railway stakeholders can provide more reliable information about stations, tracks, platforms, and operational assets [1]. However, such data may require preprocessing, cleaning, transformation, and alignment before it can be used for simulation.

Standardized formats are also important for railway data exchange. RailML, GTFS and related railway modelling standards provide structured ways to describe infrastructure, rolling stock, timetables, and operational data [8, 6]. They improve interoperability between tools, but converting raw datasets into such formats can require significant modelling effort.

Operational Data

Operational data describes the planned or actual movement of trains. It may include timetables, train identifiers, station stops, arrival and departure times, delays, platform assignments, rolling stock information, and disruption records. Operational data is essential for transforming a static infrastructure model into a dynamic simulation scenario.

Timetable data provides the planned structure of train services. It defines which trains should run, when they should depart, where they should stop, and how they should connect to other services. Punctuality data provides information about actual train movements and deviations from the planned timetable. It can be used to analyse delay patterns, validate simulation results, or generate realistic train routes.

Operational data is often difficult to use directly. It may contain missing records, inconsistent station identifiers, cancelled trains, duplicated entries, or timing errors. It may also describe movements at station level, while the simulation requires detailed paths at track level. Therefore, cleaning and transformation are necessary before operational data can be used in a simulator.

A key challenge is the alignment between operational data and infrastructure data. A train record may indicate that a train travelled from one station to another, but the infrastructure model must contain a valid path between these locations. If the infrastructure is incomplete or if station identifiers do not match, the route cannot be generated. This challenge is particularly important in the micro-level model, where route generation depends on detailed track connectivity.

2.3 Railway Simulators

Simulation plays an important role in the analysis, planning, and evaluation of railway systems. Since railway operations involve many interacting components, it is often difficult to test new strategies directly on the real network. Simulators provide a controlled environment in which infrastructure layouts, timetables, train movements, disruptions, and operational scenarios can be evaluated before being applied in practice.

In the context of DTRs, simulation is particularly important because the virtual representation must be able to reproduce at least part of the behaviour of the physical system. The required level of detail depends on the objective of the study. A high-level model may be sufficient for analysing station-to-station movements, while a detailed infrastructure model is required to study switches, platforms, signalling constraints, local conflicts, and route feasibility.

Railway simulators can be classified according to several criteria. A first distinction concerns the level of granularity. Macroscopic simulators represent traffic at an aggregated level, using stations, lines, flows, or timetable events. Microscopic simulators represent individual trains and infrastructure elements in more detail. A second distinction concerns the modelling objective: some tools focus on operational simulation, while others focus on infrastructure design, timetable planning, capacity analysis, or real-time traffic management. Finally, simulators differ in openness, extensibility, data requirements, interoperability, and support for automated workflows.

2.3.1 Overview of Existing Simulators

Several simulation platforms exist for railway and transport systems. Some are specifically designed for railways, while others are general-purpose or multimodal traffic simulation environments that can be adapted to rail use cases. Table [**?**] summarizes the main tools considered in this thesis.

Table 2.1: Comparative analysis of railway and transport simulation tools

Simulator	Type	Main focus	Main strengths	Main limitations	Relevance for this thesis
OpenTrack	Microscopic railway simulator	Railway operation, timetable evaluation, capacity and infrastructure analysis	Detailed railway-specific modelling; supports infrastructure, rolling stock, timetables and signalling constraints; provides train graphs, occupation diagrams and operational statistics	Requires detailed and well-structured railway data; less suited for fully custom open data pipelines; not primarily designed as an open experimental framework	Highly relevant as a reference railway simulator, but less suitable for the implemented data-driven and reproducible pipeline
RailSys	Professional railway planning and simulation suite	Capacity analysis, timetable planning, travel time computation, headway analysis and ETCS-related studies	Strong railway-specific capabilities; designed for infrastructure managers and railway operators; supports detailed operational and capacity studies	Commercial and professional environment; limited openness for academic reproducibility; requires detailed infrastructure and operational input data	Relevant for comparison with professional railway tools, but less aligned with the objective of building an open and programmable prototype
OSRD	Open-source railway-specific platform	Railway infrastructure design, capacity analysis, timetabling, simulation and short-term path requests	Open-source; railway-oriented by design; supports infrastructure and timetable workflows; suitable for capacity and pathfinding studies	More oriented toward railway design and planning workflows; integration with a custom heterogeneous data-processing pipeline may require additional modelling effort	Relevant because it is open-source and railway-specific, but SUMO was preferred for its flexibility, scriptability and simulation workflow control

Continued on next page

Table 2.1 – continued from previous page

Simulator	Type	Main focus	Main strengths	Main limitations	Relevance for this thesis
SUMO	Open-source microscopic traffic simulator	Multimodal traffic simulation, route generation, demand modelling, public transport and railway simulation	Open-source and highly scriptable; supports XML-based network generation, routing tools, outputs, TraCI interaction and large-scale automated workflows; includes railway-specific features such as rail edges, train stops and rail signals	Not originally designed as a railway-only simulator; detailed signalling, interlocking, dispatching and platform assignment require additional modelling effort	Selected as the main simulation platform because it enables full control over network generation, trip generation, execution and output processing
AnyLogic / MATSim	General-purpose or transport simulation platforms	Agent-based simulation, multimodal mobility, large-scale transport demand modelling	Useful for multimodal and agent-based transport scenarios; suitable for studying passenger behaviour, demand patterns or large-scale mobility systems	Limited native support for detailed railway infrastructure constraints such as switches, interlocking, block occupation and platform-level operations	Useful as general transport simulation references, but less appropriate for the railway infrastructure reconstruction objectives of this thesis

The choice of a simulator depends on the intended use case. For a railway digital twin foundation, several criteria are important. The simulator must represent the infrastructure at the required level of detail, support train route generation and execution, be compatible with available data sources, produce outputs that can be compared with operational data, and allow reproducible experimentation.

Specialized railway simulators such as OpenTrack and RailSys provide strong railway-specific capabilities. They are designed to represent timetables, train movements, signalling systems, capacity constraints, and detailed infrastructure. Their main advantage is that they directly encode railway concepts that are simplified or absent in general-purpose traffic simulators. However, they often require detailed infrastructure and timetable data, and their integration into a fully custom data-processing pipeline can be less straightforward.

OSRD is an interesting alternative because it is both open-source and railway-specific. It directly targets railway infrastructure design, capacity analysis, timetabling, and pathfinding. However, its data model is more oriented toward railway planning workflows, whereas this thesis focuses on transforming heterogeneous infrastructure and operational data into simulation-ready artefacts within a flexible and programmable environment.

SUMO occupies a different position. It is less railway-specialized than OpenTrack, RailSys, or OSRD, but it provides a highly flexible simulation environment. Its main strengths are openness, scalability, scriptability, and the availability of tools for network generation, routing, demand generation, output processing, and online interaction. These characteristics make SUMO suitable for experimental research where the objective is not only to run a simulation, but also to generate the simulation network automatically from data, modify scenarios programmatically, and analyse outputs with custom scripts.

For this thesis, SUMO was selected because it allows the complete simulation workflow to be controlled. Networks can be generated from XML files, routes can be created automatically, simulations can be executed through command-line tools, and outputs can be parsed into datasets suitable for analysis. This level of control is essential for comparing macro-level and micro-level railway representations and for evaluating the feasibility of integrating real operational data into a simulation pipeline.

2.3.2 Limitations and Gaps in Current Simulators

Existing simulators provide important capabilities, but several limitations remain when the objective is to construct a railway digital twin from heterogeneous real-world data. A first limitation concerns data integration. Many simulators assume that infrastructure, timetable, and demand inputs are already available in a consistent and simulator-compatible format. In practice, railway data often comes from multiple sources, with different identifiers, geometries, levels of detail, and temporal resolutions. Transforming these datasets into valid simulation inputs remains a major challenge.

A second limitation concerns the balance between realism and scalability. Detailed railway simulators can represent signalling, switches, platforms, and operational conflicts, but this usually requires high-quality infrastructure data and careful configuration. Simpler models are easier to generate and scale, but they omit important operational constraints. This creates a trade-off between macroscopic representations, which are easier to use with large datasets, and microscopic representations, which are more realistic but more sensitive to data quality and topological inconsistencies.

A third limitation concerns reproducibility and openness. Some railway simulation tools are commercial or rely on proprietary workflows. This can be appropriate for industrial applications, but it may limit academic reproducibility. Open-source tools such as SUMO and OSRD reduce this barrier, but they still require significant preprocessing and modelling work before real railway data can be used.

A final limitation concerns the connection between simulation and operational data. Many simulation studies focus either on infrastructure modelling or on timetable evaluation, but fewer provide an end-to-end workflow that starts from raw infrastructure and punctuality data, generates a simulation-ready railway network, executes train movements, and compares simulation outputs with real observations. This gap directly motivates the work presented in this thesis.

2.3.3 SUMO: General Features

SUMO is an open-source traffic simulation suite designed to model large-scale transport systems. It is microscopic, meaning that each vehicle is represented individually, and time-continuous, meaning that vehicle movements evolve during the simulation rather than being represented only as isolated events. SUMO supports several transport modes, including cars, buses, bicycles, pedestrians, public transport vehicles, and rail vehicles [31, 16].

A SUMO scenario is built from several input files. The network file describes the topology of the simulated environment, including nodes, edges, lanes, connections, and permissions. Route files define vehicles, vehicle types, departure times, and paths. Additional files may define stops, detectors, traffic lights, calibrators, rerouting rules, public transport stops, or other simulation objects. A configuration file then specifies which inputs must be loaded and which outputs must be generated.

SUMO also provides several tools for scenario creation and manipulation. `netconvert` converts network descriptions into SUMO network files and can import networks from different formats, including OpenStreetMap and manually defined XML files. `duarouter` computes routes from trips or origin-destination information. `randomTrips.py` generates synthetic travel demand. `netedit` provides a graphical interface for editing networks. These tools make SUMO suitable for automated simulation pipelines.

SUMO also provides a rich set of outputs. Depending on the configuration, it can generate information about completed trips, stops, vehicle trajectories, edge-based statistics, lane-based statistics, emissions, delays, and simulation summaries. These outputs are useful because they can be transformed into structured datasets and compared with real observations. In this thesis, outputs such as trip information, stop information, and summary statistics are particularly important because they allow simulated train movements to be evaluated against operational data.

Another important feature is the availability of structured outputs. SUMO can generate trip information, stop information, vehicle trajectories, edge-based statistics, lane-based statistics, emissions, summary files, and other outputs depending on the configuration. These files can be parsed after simulation and converted into analysis-ready datasets. For this thesis, this is particularly important because the simulation outputs must be compared with railway operational data.

2.3.4 SUMO Backend Architecture

SUMO follows a modular architecture built around XML input files, command-line tools, simulation executables, and optional programming interfaces. This architecture makes it possible to generate, execute, and analyse simulations automatically.

The main simulation executable, `sumo`, runs scenarios without a graphical interface and is therefore suitable for batch experiments. The graphical version, `sumo-gui`, provides visual inspection of the simulation, which is useful for debugging network geometry, train routes, stops, and conflicts. The simulation kernel loads the network, vehicles, routes, stops, and configuration files, then updates the state of each vehicle during the simulation.

Network generation is handled by dedicated tools such as `netconvert` and `netgenerate`. In the context of this thesis, `netconvert` is especially important because it transforms node and edge definitions into a complete SUMO network. This step includes the construction of lanes, connections, permissions, and network topology. For railway simulation, this conversion step is critical because routing feasibility depends on the correctness of the generated topology.

Demand generation and routing are handled by tools such as `randomTrips.py` and `duarouter`. `randomTrips.py` can generate synthetic trips according to user-defined parameters, while duarouter computes valid paths over the network. In railway scenarios, these tools can be used to generate artificial train movements or to convert reconstructed operational trips into executable routes.

SUMO also includes Python libraries such as `sumolib`, which allows SUMO networks and XML files to be inspected and manipulated programmatically. This is important for research workflows because it makes it possible to build custom preprocessing, validation, and post-processing scripts. Together with `TraCI`, these libraries provide a bridge between SUMO and external data-processing environments.

Several auxiliary tools support the simulation workflow. `netconvert` is used for network generation, `duarouter` for route computation, `randomTrips.py` for demand generation, and `netedit` for manual inspection or editing. In addition, `sumolib` provides Python utilities for reading and processing SUMO networks and output files. `TraCI`, the Traffic Control Interface, allows external programs to interact with a running simulation by retrieving vehicle states, modifying routes, changing parameters, or controlling elements during execution.

This modular design is useful for DT development because each stage of the pipeline can be automated. Infrastructure data can be transformed into nodes and edges, operational data can be transformed into train trips, routing tools can compute paths, and output files can be converted into datasets for validation.

2.3.5 SUMO For Railway Simulation

Although SUMO was originally developed as a general traffic simulator, it includes features that can be adapted to railway simulation. Rail vehicles can be represented using vehicle types and route definitions. Railway infrastructure can be encoded through edges and lanes with rail permissions. Train stops can be represented using stopping areas attached to lanes, and schedules can be defined through departure times and planned stops. SUMO also provides support for public transport lines, rail signals, crossings, and bidirectional track usage in certain modelling configurations [16, 15].

For railway use cases, SUMO offers two important advantages. First, it can represent the network at different levels of abstraction. At the macro level, stations can be represented as nodes and station-to-station links as rail edges. At the micro level, individual track segments, platforms, and switches can be represented more explicitly. Second, SUMO supports automated scenario generation. This makes it possible to generate railway simulations from data rather than manually constructing each scenario.

However, SUMO also has limitations for railway simulation. Detailed signalling, interlocking, dispatching, platform assignment, block occupation, and conflict resolution are not automatically inferred from raw infrastructure data. They must either be simplified, explicitly encoded, or controlled through additional logic. Therefore, SUMO is well suited for building a flexible simulation-oriented digital twin foundation, but it does not remove the need for careful railway modelling.

In this thesis, SUMO is used as a simulation backend rather than as a complete railway planning system. The main objective is to evaluate whether heterogeneous infrastructure and operational datasets can be transformed into executable railway simulations. SUMO is therefore appropriate because it offers full control over the generation of networks, routes, stops, configuration files, and outputs. Its limitations are also useful from a methodological perspective, because they reveal which railway-specific elements must be explicitly reconstructed before a more complete digital twin can be achieved.

2.4 Simulation Calibration

Simulation calibration is the process of adjusting model parameters so that simulation outputs reproduce observed real-world behaviour as closely as possible. In transport simulation, calibration is necessary because a simulator is never a perfect copy of reality. It relies on assumptions, simplifications, parameters, and incomplete data. Without calibration, a simulation may be technically executable while still producing unrealistic travel times, traffic flows, delays, or capacity usage.

In microscopic traffic simulation, calibration often concerns parameters related to vehicle behaviour, such as acceleration, deceleration, reaction time, lane-changing behaviour, route choice, and departure distributions. In railway simulation, the relevant parameters are different. Calibration may concern running times, dwell times, acceleration and braking behaviour, platform stopping times, route choices, dispatching rules, delay distributions, and capacity constraints.

Several calibration approaches can be used. Manual calibration relies on expert knowledge and compares simulation outputs with reference data. It is simple and interpretable, but can be subjective and difficult to scale. Optimization-based calibration defines an objective function measuring the difference between simulated and observed data, then searches for parameter values that minimize this difference. Data-driven calibration uses statistical or machine learning methods to estimate parameters directly from historical observations.

A calibration process generally requires three elements: reference data, comparable simulation outputs, and evaluation metrics. In railway applications, reference data may include planned and observed arrival times, departure times, dwell times, delays, platform usage, or track occupation. The corresponding simulation outputs can then be compared using metrics such as mean absolute error, root mean square error, travel time error, delay distribution similarity, punctuality indicators, or differences between observed and simulated traffic volumes.

SUMO includes calibration-related mechanisms, especially for road traffic. For example, calibrators can insert or remove vehicles in order to match observed flow and speed values over specific time intervals [14]. More generally, SUMO can also be integrated into external calibration workflows because simulations can be launched automatically, parameters can be modified through scripts, and outputs can be analysed programmatically.

In this thesis, calibration is not the main objective. The purpose is not to tune the simulator until it reproduces Belgian railway operations with high precision, but to construct simulation-ready railway models and evaluate whether heterogeneous infrastructure and operational data can be transformed into feasible SUMO scenarios. Nevertheless, calibration remains important for future work. Once infrastructure reconstruction and route feasibility issues are addressed, calibration could be used to adjust dwell times, running times, departure distributions, and delay behaviour so that the simulated railway system better matches historical punctuality data.

2.5 Research Gaps

The literature shows that digital twins and railway simulators provide a strong foundation for railway analysis, but several gaps remain when the objective is to construct a simulation-oriented railway digital twin from heterogeneous real-world data.

The first gap concerns data transformation. Existing simulation tools often assume that infrastructure, timetable, and operational data are already available in consistent and simulator-compatible formats. In practice, railway datasets differ in structure, identifiers, spatial precision, temporal granularity, and level of detail. Infrastructure data may describe stations, tracks, switches, platforms, or geometries, while operational data may only describe train movements at station level. Transforming these heterogeneous datasets into valid simulation inputs is therefore a central challenge.

The second gap concerns the level of network representation. Macro-level models, where stations are represented as nodes and railway connections as edges, are easier to generate and more robust for large-scale simulation. However, they ignore detailed infrastructure constraints such as platforms, switches, parallel tracks, and station-level routing. Micro-level models are closer to the physical infrastructure, but they are more sensitive to missing connections, incorrect topology, platform assignment errors, and routing inconsistencies. The practical trade-off between macro-level robustness and micro-level realism remains insufficiently explored.

The third gap concerns the integration of real operational data. Synthetic trips can test network connectivity, but they do not reproduce real railway operations. Punctuality data provides a more realistic basis, but it does not contain the exact physical route followed by each train. Train journeys must therefore be reconstructed from station sequences and mapped onto the generated network. This is manageable at the macro level, but more difficult at the micro level, where each movement must correspond to a valid sequence of track segments and platform stops.

This micro-level difficulty is especially important in this thesis. The failure to reconstruct some real train routes is not only an implementation limitation, but also a methodological finding. It shows that high infrastructure coverage does not necessarily imply route feasibility. A reconstructed network must also be topologically consistent, and the dual graph used for routing must preserve valid transitions between track segments. Missing connections, wrong directions, inaccurate platform matching, or excessive segmentation can prevent real station sequences from becoming executable routes.

This thesis addresses these gaps by developing a SUMO-based pipeline at two levels of abstraction. The macro-level model evaluates the integration of real operational data into a simplified station-to-station simulation. The micro-level model evaluates the feasibility and limitations of detailed infrastructure reconstruction based on tracks, switches, platforms, and routing structures. Together, these models make it possible to analyse the trade-off between simplicity, scalability, realism, and simulation feasibility.

Chapter 3

Proposed Methodology

This chapter presents the methodology used to construct a simulation-oriented digital twin of a railway network. The objective is not to build a complete real-time digital twin, but to define the main steps required to transform heterogeneous railway datasets into SUMO-compatible simulation models.

The proposed approach is based on two complementary levels of representation. The macro-level model represents the railway network as a station-to-station graph and is mainly used to integrate real operational data into simulation. The micro-level model represents the infrastructure in more detail through tracks, switches, platforms, and routing structures. This distinction makes it possible to compare a simplified and robust representation with a more detailed but more demanding one.

3.1 Positioning of This Work

The state of the art shows that railway digital twins and simulation tools provide useful foundations for analysing railway systems. However, constructing an executable simulation from real railway data remains difficult. Infrastructure data and operational data are often heterogeneous, incomplete, and represented at different levels of detail. Existing simulators also generally assume that the required input files are already available in a consistent and simulator-compatible format.

This thesis addresses this gap by focusing on the transformation of railway data into simulation-ready artefacts. The contribution is therefore not the development of a new simulator, but the design and evaluation of a data-driven pipeline that connects infrastructure reconstruction, operational data integration, route generation, simulation execution, and output analysis.

The work is positioned around the comparison of two modelling levels. The macro-level model evaluates how far real punctuality data can be integrated into a simplified station-to-station railway simulation. The micro-level model investigates whether a more detailed infrastructure representation can support realistic railway routing based on tracks, platforms, switches, and a reconstructed dual graph. This comparison is central to the thesis because it highlights the trade-off between abstraction, scalability, infrastructure realism, and simulation feasibility.

3.2 Framework

The proposed framework follows a data-driven pipeline that converts raw railway datasets into SUMO-compatible simulation scenarios. The pipeline is composed of four main stages: data preprocessing, infrastructure modelling, trip generation, and simulation evaluation.

First, the available railway datasets are cleaned and standardized. Infrastructure and operational data must be aligned because they do not always share the same identifiers, formats, or level of detail. This step includes filtering the selected simulation area, harmonizing station information, processing geographical coordinates, and converting temporal values into simulation-compatible timestamps.

Second, the railway infrastructure is reconstructed according to the selected level of abstraction. At the macro level, stations are represented as nodes and railway links as edges. This graph is suitable for reconstructing station-to-station train movements from punctuality records. At the micro level, the infrastructure is represented through individual track segments, detected switches, platform-track associations, and routing structures. This representation is closer to the physical infrastructure, but it requires stronger topological consistency.

Third, train movements are generated. In the macro-level model, trips are reconstructed from real punctuality records by grouping train passages by train identifier and operating day, then sorting them chronologically. In the micro-level model, two strategies are considered: synthetic trip generation and real-data-based trip reconstruction. Synthetic trips are used to test network connectivity, while real-data trips aim to reproduce observed train movements on the detailed infrastructure.

Finally, the generated networks and routes are executed in SUMO. The simulation outputs are then processed into structured datasets that can be analysed and compared with the expected behaviour of the model. This evaluation focuses on the quality of the generated infrastructure, the feasibility of train routing, and the ability of each representation to support railway simulation.

3.3 Why SUMO?

SUMO was selected because the objective of this thesis requires a flexible, open-source, and scriptable simulation platform. Specialized railway simulators such as OpenTrack, RailSys, or OSRD provide strong railway-specific functionalities, but they often require detailed and well-structured railway input data. In contrast, this work focuses on building a custom pipeline from heterogeneous datasets, which requires full control over network generation, route generation, execution, and output processing.

SUMO was selected because it is open-source, scriptable, and compatible with automated data-processing workflows. Its XML-based input structure makes it possible to generate networks, routes, stops, and configuration files directly from Python scripts. It also provides command-line tools which are useful for converting networks, generating trips, and computing executable routes.

SUMO is well suited to this objective because its XML-based structure allows networks, routes, stops, vehicle types, and configuration files to be generated automatically from Python scripts. It also provides command-line tools such as `netconvert`, `randomTrips.py`, and `duarouter`, which support automated network conversion, synthetic trip generation, and route computation.

Another advantage is that SUMO can support both levels of representation considered in this thesis. At the macro level, stations can be represented as nodes and station-to-station connections as railway edges. At the micro level, more detailed infrastructure can be represented through rail edges, lanes, train stops, and route definitions. This flexibility makes SUMO suitable for comparing simplified and detailed railway simulation approaches within the same environment.

However, SUMO is not a railway-only simulator. Detailed signalling, interlocking, dispatching, platform assignment, and conflict management are not automatically inferred from raw data. These limitations must be taken into account when interpreting the simulations. For this thesis, this trade-off is acceptable because the main objective is to study how railway data can be transformed into simulation-ready models, rather than to reproduce all operational rules of a professional railway simulator.

3.4 Data Available

The methodology relies on two main categories of data: infrastructure data and operational data. Infrastructure data describes the railway network itself, while operational data describes how trains move through this network. Both are required to construct a simulation-oriented railway digital twin.

The available data is not directly usable by SUMO. It must first be cleaned, filtered, transformed, and converted into simulation files. The type of processing required depends on the level of representation. The macro-level model mainly uses stations and station-to-station links, whereas the micro-level model requires track geometry, platforms, switches, and routing structures. The data used in this thesis comes from Infrabel Open Data Portal And DTSC who has access to private data directly from Infrabel.

3.4.1 Infrastructure Data

Infrastructure data describes the static railway network. It is used to generate the physical or topological environment in which trains circulate. The macro-level and micro-level models use different infrastructure datasets because they represent the railway system at different granularities.

At the macro level, the network is represented as a graph of stations and direct connections. This abstraction is sufficient for simulating train movements between neighbouring stations, but it does not include platforms, switches, signals, or detailed station layouts.

Dataset	Example attributes	Reference value	Role in Methodology
Stations	Station identifier, station name, coordinates	559 stations	Create SUMO nodes
Stations Connections	Origin, destination, distance, geometry	1385 tracks	Create SUMO railway edges
Selected Simulated Area	List of neighbouring stations	User-defined	Restrict the simulation scope

Table 3.1: Macro-Level Infrastructure Data Overview

At the micro level, the infrastructure is represented in more detail. Track segments are used to build the railway graph, switches are inferred from the topology, and platforms are associated with nearby track segments. This representation is necessary for more realistic routing, but it is more sensitive to missing connections or incorrect topology.

Dataset	Example attributes	Reference value	Role in Methodology
Tracks segments	Track identifier, geometry, length	25068 tracks	Build detailed railway edges
Stations	Station identifier, station name, coordinates	554 stations	Support platform detection
Platforms	Station name, platform number, station's position	1560 platforms	Create SUMO train stops
Switches	Node identifier, coordinates, connected tracks	10379 switches	Support routing and topology analysis

Table 3.2: Micro-Level Infrastructure Data Overview

3.4.2 Operational Data

Operational data describes train movements over the railway network. In this thesis, this information is contained in a single punctuality dataset. Each record corresponds to the passage of a train at a station and includes information related to train identification, station passage, planned times, observed times, and delays.

This dataset is used to reconstruct ordered train journeys. For each train and operating day, station records are grouped and sorted chronologically. At the macro level, these sequences can be mapped onto the station-to-station graph. At the micro level, they must be mapped onto a detailed path composed of tracks, platforms, and valid connections, which makes the process more complex.

Dataset	Main attribute groups	Example attributes	Reference value	Role
Punctuality dataset	Train identification, station passage, timetable information, observed operation, delays	Train identifier, operating date, station identifier, planned arrival/departure time, observed arrival/departure time, arrival delay, departure delay	Depends on selected time window	Reconstruct real train journeys and generate simulation routes

Table 3.3: Operational Data Overview

Overall, the available data provides the basis for constructing both macro-level and micro-level railway simulations. The infrastructure data defines the network to be simulated, while the operational data defines the train movements to reproduce. The central methodological challenge is therefore to align these datasets at the appropriate level of detail and convert them into valid SUMO inputs.

Chapter 4

Implementation

4.1 Macro-Level Modeling

The macro-level model constitutes the first operational implementation of the simulation-oriented railway digital twin developed in this thesis. Its objective is to generate a simplified but executable representation of the Belgian railway network in SUMO. At this level of abstraction, stations are represented as nodes and railway connections between stations as edges. Detailed infrastructure elements such as platforms, switches, individual tracks, signals, and station-level routing constraints are intentionally omitted.

This abstraction makes the model computationally manageable and suitable for validating the complete workflow, from raw railway data to simulation outputs. It also provides a baseline against which the more detailed micro-level model can later be compared. The implementation was developed in Python, using Apache Spark for data processing and SUMO for simulation execution. The workflow consists of four main stages: data processing, network generation, trip generation, and simulation.

4.1.1 Data Processing Pipeline

The macro-level model starts with a data processing pipeline that transforms infrastructure and operational datasets into a format compatible with SUMO. Three main types of data are used: station data, station-to-station connectivity data, and punctuality data. Station data provides identifiers, names, and geographical coordinates. Connectivity data describes the railway links between neighbouring stations. Punctuality data contains operational records of train passages through stations.

The preprocessing follows an Extract-Transform-Load approach. First, the raw datasets are loaded into Apache Spark dataframes. Spark is used mainly because the punctuality dataset can be large and requires efficient filtering, grouping, and temporal transformations. The transformation step then standardizes identifiers, processes geographical information, removes inconsistent records when necessary, and prepares the data for simulation generation.

A filtering step is also applied to restrict the simulation to the selected area and time period. The user can choose a subset of stations to simulate, and neighbouring stations required to preserve connectivity may be included automatically. This makes it possible to focus on coherent station chains rather than the entire national network, which reduces complexity while preserving meaningful train movements.

The output of this pipeline is a set of cleaned and harmonized datasets containing station information, station-to-station links, and train circulation records. These datasets are then used to generate the SUMO network and the corresponding train routes.

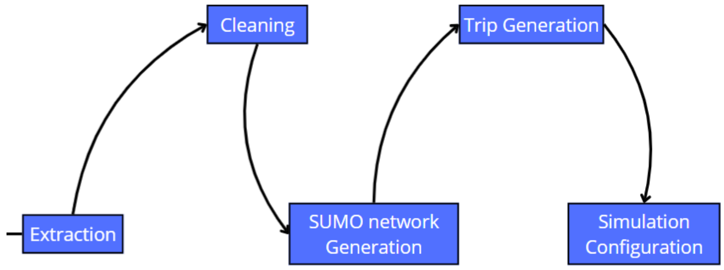

Figure 4.1: Macro-Level Data Processing Pipeline

4.1.2 Railway Network Generation

After preprocessing, the railway network is generated as a station-to-station graph. Each station becomes a node, and each direct railway connection becomes an edge. The graph can be defined as $G = (V, E)$, where V is the set of stations and E is the set of railway links between them.

For each station, the implementation stores the station identifier, name, and geographical coordinates. For each connection, it stores the origin station, destination station, distance, and, when available, the geometry of the railway corridor. Preserving the geometry allows the generated SUMO network to remain visually close to the real railway layout, while keeping the model much simpler than the actual infrastructure.

The graph is then converted into SUMO-compatible XML files. The node file defines the stations included in the simulation, while the edge file defines the rail links between them. Additional files define train stops, vehicle types, route information, and the simulation configuration. The final network is generated with netconvert, which transforms the node and edge definitions into a SUMO network file.

This representation is simple, robust, and easy to generate. However, because each station is represented as a single node, the model cannot reproduce internal station operations such as platform allocation, switch occupation, signal interactions, or local route conflicts.

4.1.3 Trip Generation

Once the macro-level network has been generated, train movements are reconstructed from the punctuality dataset. The objective is to use real operational data rather than relying only on synthetic traffic.

The punctuality dataset contains station-level records for train passages. Each record includes a train identifier, a station identifier, and temporal information such as planned and observed times. However, the dataset does not directly provide complete SUMO routes. A reconstruction step is therefore required.

For each train and operating day, the records are grouped and sorted chronologically according to the planned timetable. Consecutive station pairs are then extracted to reconstruct the sequence of stations visited by the train. These station sequences are mapped onto the macro-level network and converted into SUMO route definitions.

Algorithm 1: Macro-level railway network generation

Input: Set of railway stations S;
Set of station-to-station connections C;
Set of stations selected by the user S_{selected}.
Output: Generated SUMO network files.

Initialize an empty set of nodes $N \leftarrow \emptyset$;
Initialize an empty set of edges $E \leftarrow \emptyset$;
foreach *station* $s \in S$ **do**
  if $s \in S_{\text{selected}}$ ***or*** *s is required to preserve network connectivity* **then**
    Create a SUMO node using the identifier and coordinates of s;
    Add the generated node to N;
foreach *connection* $c \in C$ **do**
  $s_{\text{origin}} \leftarrow c$.departure_station;
  $s_{\text{destination}} \leftarrow c$.arrival_station;
  if $s_{\text{origin}} \in N$ ***and*** $s_{\text{destination}} \in N$ **then**
    Create a SUMO edge from s_{origin} to $s_{\text{destination}}$;
    Assign the distance and railway geometry of c to the edge;
    Add the generated edge to E;
Export N as a SUMO node file;
Export E as a SUMO edge file;
Generate the additional SUMO files for railway stops and configuration;
Run `netconvert` to generate the final SUMO network;
return *Generated SUMO network files*;

Departure times must also be converted into SUMO simulation time. Since SUMO represents time as seconds elapsed since the start of the simulation, each planned departure time is transformed by computing its difference from the selected simulation start date.

This approach preserves realistic train movements, station usage patterns, and departure time distributions. It therefore provides a more representative simulation scenario than purely random traffic generation, while remaining compatible with the simplified station-to-station model.

4.1.4 Simulation

After the network and routes have been generated, the macro-level scenario can be executed in SUMO. The simulation configuration references all required files: the network file, route file, additional files for stops and trains, and output definitions.

The macro-level simulation is not intended to reproduce the full Belgian railway network in a single scenario. Although the generated network can theoretically include a large number of stations, simulating the complete national traffic would introduce a high level of complexity in capacity management in frequently visited stations (stations with dense traffic, multiple platforms, many intersecting services, and complex internal routing would require infrastructure details that are not represented at this level of abstraction). For this reason, the simulation is restricted to selected parts of the network (chosen by the user). These parts correspond to chains of neighbouring stations where the train traffic remains moderate. In practice, this means that the user selects a set of stations to be

Algorithm 2: Macro-level trip reconstruction from operational data

Input: Punctuality dataset P;
Simulation start date d_{start};
Simulation duration $T_{\text{simulation}}$.
Output: SUMO route file containing the reconstructed train routes.

Filter P according to d_{start} and $T_{\text{simulation}}$;
Remove incomplete or inconsistent records from P;
Initialize an empty set of train routes $\mathcal{R} \leftarrow \emptyset$;
Group the records in P by train identifier and operating day;
foreach *train group* G **do**
    Sort the station records in G by planned departure time;
    Initialize an empty route $R \leftarrow \emptyset$;
    foreach *consecutive station pair* (s_i, s_j) *in* G **do**
        if s_i *and* s_j *are connected in the macro-level network* **then**
            if R *is empty* **then**
                Add s_i to R;
            Add s_j to R;
    if R *contains a valid sequence of stations* **then**
        Convert the first planned departure time into SUMO simulation time $t_{\text{departure}}$;
        Create a SUMO train route using R and $t_{\text{departure}}$;
        Add the generated train route to $\mathcal{R}$;
Export $\mathcal{R}$ as a SUMO route file;
return *Generated SUMO route file*;

included in the simulation, and the generation pipeline constructs the corresponding local railway subnetwork.

By focusing instead on station chains with lower traffic intensity, the simulation remains coherent with the assumptions of the macro-level model. The simplified representation is therefore used in a context where local infrastructure conflicts are less dominant and where station-to-station train movements can still be meaningfully simulated.

The duration of the simulation is configurable. The user can specify the number of operating days to be included in the scenario. This parameter determines the time window used when filtering the operational data and therefore controls the number of train movements inserted into the simulation. A short simulation period can be used for debugging and validation, while a longer period can be selected to analyse traffic behaviour over multiple days.

Once the network, train routes, stop definitions, and simulation configuration files have been generated, the simulation can be launched directly in SUMO. During execution, trains circulate along the station-to-station graph according to the reconstructed routes and departure times. Since detailed infrastructure is not represented, train movements are mainly governed by the generated routes, travel times, and SUMO's internal simulation rules.

Several outputs are produced by the simulation. The `tripinfo` file provides information about completed train journeys, the `stopinfo` file records stopping events at stations, and `summary` outputs describe the global evolution of the simulation. A post-processing module parses these XML outputs and converts them into CSV files whose structure is closer to the original punctuality data.

This post-processing step reconstructs simulated arrival and departure information for each train and station. As a result, the macro-level simulation can be compared with historical operational data and used to evaluate whether the generated digital twin reproduces station-to-station train movements in a coherent way.

The macro-level workflow can therefore be summarized as follows: raw data is cleaned and standardized, a station-to-station SUMO network is generated, real punctuality records are transformed into train routes, the scenario is simulated, and the outputs are converted into datasets suitable for validation.

4.1.5 Limitations

Although the macro-level model validates the overall pipeline, it remains limited by its level of abstraction. Stations are represented as single nodes and railway corridors as direct edges. As a result, detailed infrastructure elements such as platforms, switches, signals, and individual tracks are not explicitly represented. The model therefore cannot accurately reproduce local operational constraints, platform conflicts, switch occupation, or station-level routing decisions.

The model is also dependent on the quality of the available infrastructure and operational data. It can reconstruct train movements at station level, but it cannot identify the exact physical route followed by trains inside complex railway areas. Nevertheless, the macro-level implementation remains an important first step: it demonstrates that real operational data can be integrated into a SUMO-based railway simulation and provides a robust baseline for comparison with the micro-level model.

4.2 Micro-Level Modeling

The micro-level model was developed to overcome the main abstraction of the macro-level representation. Instead of modelling the railway network only as stations connected by aggregated links, the micro-level model aims to represent the physical infrastructure in greater detail. Tracks, platforms, switches, and operational segments are explicitly reconstructed and converted into SUMO-compatible elements.

The objective of this model is to evaluate whether a more realistic railway simulation can be generated from the available infrastructure and operational datasets. Compared with the macro-level model, this approach provides a richer representation of the railway network, but it also introduces stronger requirements on data quality and topological consistency. A valid simulation requires not only that infrastructure elements are detected, but also that they form a connected and routable network.

This approach significantly increases the realism of the generated digital twin. The explicit representation of railway assets enables more accurate train movements, better infrastructure utilization, and the possibility of evaluating local operational phenomena such as conflicts at junctions, platform occupancy, and the propagation of delays within complex stations.

The implementation follows five main stages: data processing, infrastructure modelling, trip generation, simulation, and limitation analysis.

4.2.1 Data Processing Pipeline

The micro-level data processing pipeline follows an ETL process. The raw infrastructure data contains geographical track geometries represented as sequences of coordinates. These geometries are first cleaned and normalized in order to obtain a consistent spatial representation of the railway network.

Several processing steps are then applied to enrich the infrastructure. First, switches are detected from the topology of the track network. Then, platforms are associated with nearby track segments using a spatial proximity-based method. When several stations or platforms are assigned to the same track segment, a segmentation step is applied to increase the granularity of the network. Finally, the processed infrastructure is converted into graph structures used for route generation and simulation.

The output of the pipeline consists of a detailed railway network containing track segments, detected switch-like nodes, platform-to-track associations, and routing structures. These elements are then exported into SUMO files and used to generate train movements.

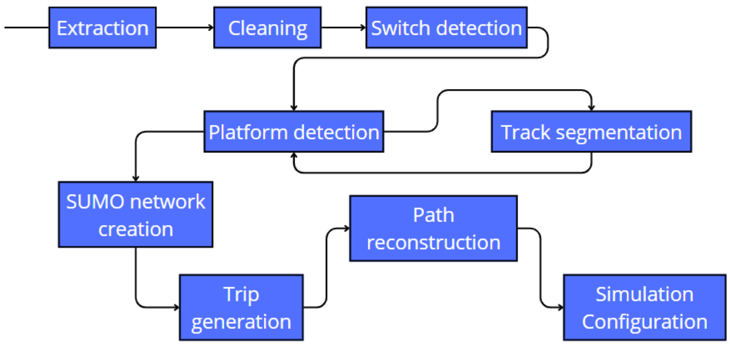

Figure 4.2: Micro-Level Data Processing Pipeline

4.2.2 Infrastructure Modeling

Switch Detection

Switch detection is based on the topology of the railway graph. Track extremities are represented as nodes, and track sections are represented as edges. For each node, the number of connected track segments is computed.

Figure 4.3: Switch detection initial setup

Nodes connected to two track segments are considered part of a continuous railway line. Nodes connected to three or more track segments are classified as switch-like elements because they represent branching or merging points in the infrastructure. These detected switches are then used to enrich the micro-level network and support routing through the reconstructed topology.

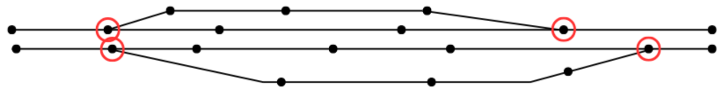

Figure 4.4: Switch detection detection part

The resulting switch and edge datasets forms the foundation of the micro-level network model and is subsequently used for graph construction, route generation, and simulation.

Figure 4.5: Switch detection final result

This approach does not require explicit switch labels in the source data. However, it depends strongly on the quality of the reconstructed track topology. Missing, duplicated, or incorrectly connected track segments can affect the number of detected switches.

Platform Detection

Platform detection aims to associate station platforms with the track segments on which trains can stop. The available data provides station locations and platform information, but these elements are not directly linked to SUMO lanes. A matching procedure is therefore required.

The implemented method relies on spatial proximity. For each station, a search area is defined around its coordinates. Nearby track segments are identified and ranked according to their distance from the station. If a station has N platforms, the N closest candidate track segments are selected and associated with the station platforms.

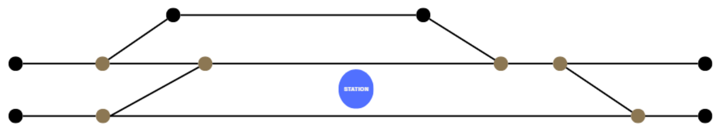

Figure 4.6: Example of platform detection initial setup

For each station, a research zone is defined around its coordinates. This research zone is a circular area with an adaptative radius (e.g., 100 meters) that encompasses the area where they are likely located. The radius of the research zone can be adjusted based on the density of platforms in the area, ensuring that it is large enough to capture all relevant track segments while minimizing the inclusion of irrelevant segments.

Within the research zone, the script identifies all track segments from the track dataset that intersect with the zone. Then, the script ranks the track segments based on their proximity to the station coordinates. The remaining track segments are classified into three categories (given a station that has n platforms):

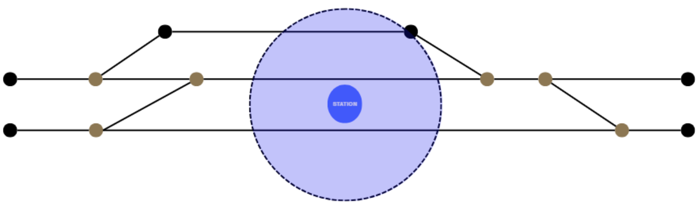

Figure 4.7: Example of platform detection searching zone

1. Best candidates: the track segments that are closest to the station coordinates and are likely to correspond to a platform ($rank \leq n$).
2. Mediocre candidates: the track segments that are within the research zone (or close to it) but are not the best candidate. These segments may correspond to additional platforms or other track segments near the station ($n \leq rank \leq 1.5n$).
3. Irrelevant segments: the track segments that are outside the research zone or are too far from the station coordinates. These segments are unlikely to correspond to platforms and can be disregarded in the platform detection process ($1.5n \leq rank \leq 2n$).

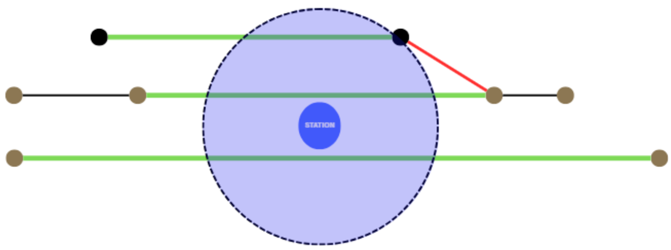

Figure 4.8: Example of platform detection ranking when the number of platform is 3

This method is simple and scalable, but it remains an approximation. It works well when platforms are located close to the corresponding tracks, but it can be less reliable in complex stations where many parallel tracks are located close to each other.

Track Segmentation

Track segmentation is introduced to improve the granularity of the reconstructed infrastructure. In some cases, a single track segment may be associated with several stations or platforms. This creates ambiguity because the same segment cannot represent several distinct stopping locations accurately.

To address this issue, long or conflicting track segments are split into smaller sub-segments while preserving the original geometry. After segmentation, the platform detection process can be applied again on a more detailed infrastructure representation. This improves the association between platforms and tracks and produces a network better suited for simulation.

First, all segments assigned to more than one station are identified. Only these conflicting segments are processed, while the remaining network remains unchanged. Each selected segment is then divided into smaller sub-segments according to a maximum segment length threshold of 800 meters.

Figure 4.9: Track segmentation Initial setup

New sub-segments are created exclusively using the coordinates already present in the original polyline representation. No interpolation or artificial points are introduced. Distances are computed after projecting the geographical coordinates into the Belgian Lambert 72 coordinate system (EPSG:31370), allowing accurate metric measurements.

For each generated sub-segment, new extremity nodes are created when necessary and the corresponding track attributes are recomputed. The resulting infrastructure therefore preserves the original network geometry while providing a finer representation of the railway infrastructure.

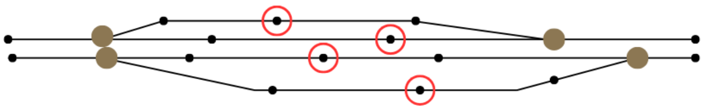

Figure 4.10: Track segmentation applied to a conflicting segment

After segmentation, the platform matching procedure can be repeated on the newly generated segments. The process continues until each station platform is associated with a distinct segment. In practice, a single segmentation iteration was sufficient to eliminate all observed conflicts in the Belgian railway network dataset.

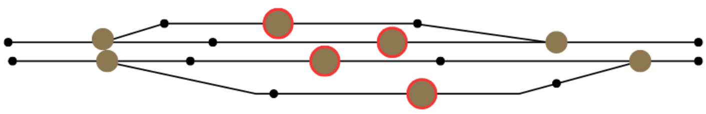

Figure 4.11: Track segmentation results

This approach allows the model to maintain a realistic infrastructure representation while ensuring that stations and platforms can be uniquely mapped to simulation entities. Such a requirement is particularly important for subsequent routing, timetable generation, and railway traffic simulation tasks.

Dual Graph Construction

A dual graph is constructed to support route computation at the micro level. In this representation, each track segment becomes a node of the graph, and each feasible transition between two consecutive track segments becomes a directed edge.

This representation makes it possible to compute routes between platforms rather than only between stations. Dijkstra's shortest-path algorithm is then applied to find the shortest valid path through the dual graph. The resulting sequence of track segments is converted into SUMO edge identifiers and inserted into route definitions.

A route cache is used to avoid recomputing identical paths. Each computed route is stored according to its origin station, destination station, and platform pair. This reduces the computational cost when several trains use the same station-to-station movement.

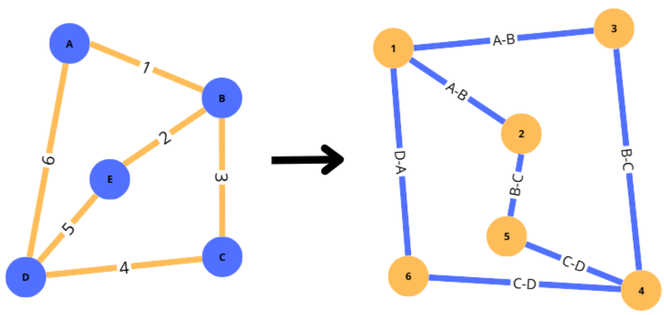

Figure 4.12: Dual graph

The construction of the dual graph is performed in two stages.First, each track segment is associated with its start and end junctions. A temporary lookup structure is generated in order to quickly retrieve all tracks connected to a given infrastructure node. This preprocessing step accelerates the graph construction by avoiding repeated searches through the complete infrastructure dataset.Next, the directed graph is created using the `NetworkX` library. For every pair of track segments, a directed edge is added whenever the destination node of the first track corresponds to the origin node of the second track. Formally, given two track segments (t_i) and (t_j), a directed edge is created if

$$\text{end}(t_i) = \text{start}(t_j). \tag{4.1}$$

The resulting edge represents a valid train movement from one track segment to the next. Each edge is assigned a weight corresponding to the physical length of the originating track segment. Consequently, shortest-path computations minimize the travelled distance across the railway infrastructure.

The resulting graph can be formally defined as $G = (V, E)$,where each vertex $(v \in V)$ represents a railway track segment and each directed edge $((u, v) \in E)$ denotes a feasible transition between two consecutive track segments.

Finally, whenever the operational dataset contains trips between station pairs that have not yet been processed, the corresponding routes are computed on demand and added to the cache. This strategy guarantees that every station-to-station movement required by the timetable is associated with a valid infrastructure path before the simulation is generated.

The dual graph is therefore a central component of the micro-level model. It provides the link between detailed infrastructure reconstruction and operational route generation. However, it also becomes a critical point of failure: if the graph is incomplete or incorrectly connected, real train journeys cannot be mapped onto the reconstructed infrastructure.

4.2.3 Trip Generation

Trip generation defines the train movements executed in the micro-level simulation. Two strategies are considered: random trip generation and real-data-based trip generation.

Random Trips

The random-trip approach generates synthetic train movements on the reconstructed railway network. Its objective is not to reproduce a real timetable, but to test whether the generated infrastructure is connected and usable in SUMO.

Candidate departure and arrival locations are selected from valid station platforms. Weighting files are generated so that SUMO's `randomTrips.py` selects only railway platforms as origins and destinations. Intermediate tracks, switches, and non-operational sections are excluded from the selection process. Additional parameters, such as insertion rate, minimum trip distance, and simulation duration, are used to control the density and realism of the generated traffic.

After trip generation, `duarouter` is used to compute executable routes through the SUMO network. This approach provides a practical way to validate the micro-level infrastructure and verify whether trains can circulate through the reconstructed network.

Real Data Trips

The real-data trip generation strategy aims to reproduce train movements from the punctuality dataset. The operational records are first cleaned and grouped by train identifier and operating day. For each train, station records are sorted according to the planned timetable in order to reconstruct an ordered sequence of stops.

The main difficulty is that each movement between two consecutive stations must be mapped onto a valid path in the detailed infrastructure. For this reason, each station must first be associated with one or more detected platforms. Then, for every pair of consecutive stations, the dual graph is used to compute a path between the selected origin and destination platforms.

Conceptually, this approach is closer to real railway operations than random trip generation because it preserves the structure of actual train services. However, in the current implementation, it did not produce a complete exploitable simulation. Many consecutive station pairs could not be connected because no valid path was found in the generated dual graph. As a result, majority of real train services had to be rejected before route generation.

This issue is an important methodological finding. It shows that high infrastructure coverage is not sufficient to guarantee simulation feasibility. At the micro level, the network must also be topologically consistent and routable. Missing connections, incorrect orientations, or imperfect platform assignments can prevent a route from being reconstructed even when the corresponding movement exists in the real railway network.

4.2.4 Simulation

Once the micro-level infrastructure and trips have been generated, the scenario can be executed in SUMO. The simulation uses detailed railway edges, train stops, route files, vehicle definitions, and configuration files.

Two simulation scenarios are considered. The first one uses random trips. This scenario is mainly used to test the generated network under controlled conditions and to verify that trains can be inserted, routed, and simulated on the reconstructed infrastructure. It provides a useful validation step because it tests the physical connectivity of the generated network without depending on the completeness of real operational routing.

The second scenario attempts to use real-data trips reconstructed from the punctuality dataset. This scenario is more realistic because it aims to reproduce observed railway operations. However, it is also more demanding because every train service must be mapped onto valid platform-to-platform paths in the dual graph. In the current implementation, this requirement could not always be satisfied, preventing the generation of a complete real-data-based micro-level simulation.

The random-trip simulation therefore demonstrates that the generated infrastructure can support train movement in SUMO, while the real-data simulation highlights the remaining difficulty of aligning real operational trajectories with a detailed reconstructed infrastructure.

4.2.5 Limitations

The micro-level model provides a more detailed representation of the railway infrastructure, but it is also more sensitive to data quality and reconstruction errors. Missing connections, incorrect track orientations, imperfect switch detection, or inaccurate platform assignments can prevent valid routes from being computed. These issues are less visible at the macro level, but they become critical when trains must follow continuous physical paths through the network.

The main limitation concerns the integration of real operational data. Random trips can be generated and simulated more easily because they only require feasible routes between selected platforms. Real-data trips are more constrained because each consecutive station pair in the timetable must correspond to a valid path in the dual graph. In the current implementation, this condition was not always satisfied, which prevented the construction of a fully exploitable real-data micro-level simulation.

This limitation does not invalidate the micro-level approach. On the contrary, it reveals an important requirement for railway digital twin construction: a detailed infrastructure model must be evaluated not only by the number of detected elements, but also by its ability to support operational routing. The micro-level model therefore provides both a more realistic infrastructure representation and a clear identification of the challenges that must be solved before real-data-based detailed railway simulation can be achieved.

Chapter 5

Results

This chapter presents the results obtained from the modeling and simulation pipeline developed in this thesis. The evaluation focuses on two main aspects. First, the generated datasets are compared with the available reference data in order to assess their completeness. Second, the infrastructure elements extracted during the modeling process are analysed to evaluate the quality of the generated railway representation.

The results are presented separately for the macro-level and micro-level approaches when necessary. This distinction is important because the two models do not represent the railway network at the same level of detail.

5.1 Dataset Validation

Dataset validation evaluates the completeness of the generated railway data by comparing it with the available reference datasets. This step focuses on the main infrastructure elements required by the macro-level and micro-level models: stations, platforms, and tracks. The objective is not yet to evaluate the operational realism of the simulation, but to verify whether the generated datasets contain the elements needed for the rest of the modeling pipeline.

5.1.1 Station Coverage

Station coverage measures the number of stations detected and integrated into the generated datasets. This metric is important because stations are used as graph nodes in the macro-level model and as reference points for platform and track association in the micro-level model.

Simulation Level	Generated Data	Reference Data	Coverage (%)
Macro-Level	559 Stations	559 Stations	100
Micro-Level	553 Stations	554 Stations	99.82

Table 5.1: Station coverage for macro-level and micro-level simulations

The results show that station coverage is almost complete for both modeling approaches. The macro-level model preserves all stations from the reference dataset, while the micro-level model misses only one station. This confirms that both generated datasets provide a reliable station basis for the subsequent infrastructure modeling and simulation steps.

5.1.2 Platform and Track Coverage

The second validation step concerns platforms and tracks. These elements are especially important for the micro-level model, since platforms define possible stopping locations and tracks define the physical railway segments used for circulation.

Category	Reference Data	Simulation Data	Coverage (%)
Platforms	1560 platforms	1551 platforms	99.42
Tracks (Macro-Level)	1385 tracks	1385 tracks	100
Tracks (Micro-Level)	25028 tracks	26721 tracks	106.59

Table 5.2: Platform and track coverage for micro-level simulation

Platform coverage is high, with 1551 generated platforms out of 1560 reference platforms. This indicates that the platform detection and matching process is able to recover almost all platform elements required for the micro-level simulation.

Track coverage must be interpreted differently depending on the modeling level. At the macro level, the generated network reproduces all 1385 station-to-station connections from the reference data. At the micro level, the number of generated tracks is higher than the reference value. This does not necessarily indicate an error, since the micro-level model transforms the infrastructure into simulation-ready segments. As a result, some reference tracks may be split into several SUMO edges during preprocessing, switch detection, and track segmentation.

Overall, the dataset validation confirms that the generated infrastructure is highly complete. However, these coverage metrics only evaluate the presence of infrastructure elements. They do not yet guarantee that platforms are correctly positioned or that all generated tracks form valid routes for simulation, which is analysed in the following sections.

5.2 Infrastructure Modeling Results

After validating the general completeness of the generated datasets, the next step is to evaluate the infrastructure modeling process itself. This section focuses on the elements reconstructed or inferred during implementation, especially platforms and switches, which are essential for the micro-level representation of the railway network.

The results confirm that the generated infrastructure provides a strong basis for simulation. Stations are almost completely preserved in both modeling approaches, and most platforms are successfully matched in the micro-level model. Switch detection requires a more careful interpretation, because switches are inferred from the topology of the generated network rather than directly copied from the reference dataset.

Category	Reference Data	Simulation Data	Detection (%)
Stations (Macro-Level)	559	559	100
Stations (Micro-Level)	554	553	99.82
Platforms (Micro-Level)	1560	1551	99.42
Switches (Micro-Level)	10379	12343	118.92

Table 5.3: Detected stations, platforms and switches for macro-level and micro-level simulations

Switch detection provides a different type of result. The reference data contains 10379 switches, while the generated micro-level infrastructure contains 12343 detected switches, corresponding to a coverage of 118.92%. As with micro-level track coverage, this value above 100% must be interpreted carefully. It does not simply mean that the model detects more correct switches than expected. Instead, it indicates that the algorithm identifies more candidate switch points than the number of switches available in the reference dataset.

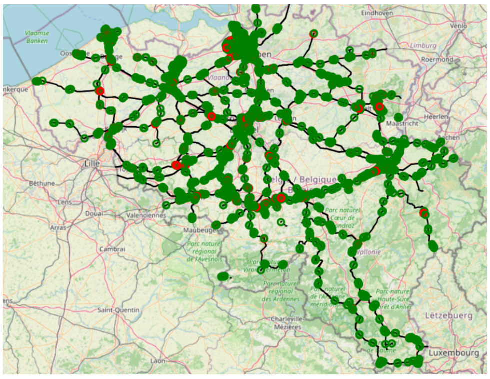

Figure 5.1: Network before and after switch detection (Green dots represent detected switches with 3 connections and Red dots represent detected switches with 4 or more connections)

The visual comparison before and after switch detection illustrates this effect. The detected switches enrich the micro-level infrastructure by making branching points explicit and by improving the network representation for routing. In dense areas, such as Bruxelles-Midi, the method provides a more detailed description of the railway topology, while also showing the difficulty of matching generated topological elements exactly with reference infrastructure objects.

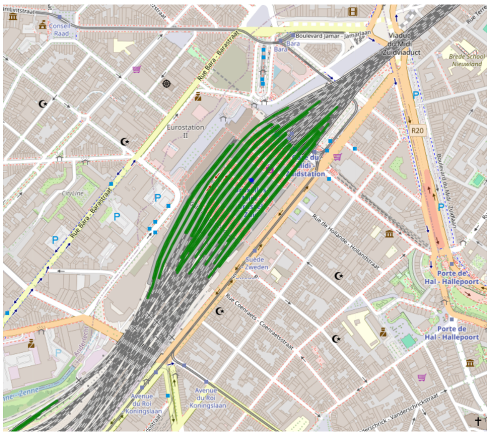

Figure 5.2: Platform matching result on Bruxelles-Midi

Platform matching provides another qualitative validation of the generated infrastructure. The matching figures show how station platforms are associated with nearby railway tracks and make it possible to inspect whether the selected track segments are plausible stopping locations. The results show that the method performs well in many cases, but that its quality depends on the local station configuration.

Figure 5.3: Platform matching quality map with examples

The platform matching quality examples illustrate this variability. Some stations, such as Tubize, show good-quality matches with small distances between platforms and candidate tracks. Other stations, such as Weerde, correspond to intermediate cases where the match remains usable but less precise. More difficult cases, such as Hansbeke, show that the algorithm can fail when the search area is larger or when the surrounding track configuration does not provide a clear candidate.

Overall, the infrastructure modeling results confirm the value of the micro-level approach. Switch detection and platform matching make it possible to transform raw infrastructure data into simulation artefacts. However, these results also show that high coverage does not automatically guarantee perfect operational correctness. Local topology, platform-track association, and route feasibility must still be evaluated through simulation.

5.3 Simulation Results

This section evaluates whether the generated railway models can be executed in SUMO. The macro-level simulation is evaluated using real operational data on a selected chain of neighbouring stations, while the micro-level simulation is evaluated using random trips on the reconstructed detailed network. These two scenarios illustrate the current simulation capabilities of the proposed framework and the different roles of the two levels of representation.

5.3.1 Macro-Level Simulation Results

The macro-level simulation was successfully executed on a selected four-station chain: Cour-sur-Heure, Ham-sur-Heure, Berzée, and Pry. This area was chosen because it forms a coherent local railway subnetwork with moderate traffic and a simple station-to-station structure.

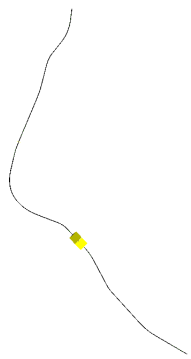

Figure 5.4: Macro-level simulation network for the selected four-station chain

In this scenario, train movements are reconstructed from the punctuality dataset and converted into SUMO routes. The generated traffic therefore corresponds to real operational movements rather than synthetic trips. This validates the ability of the macro-level pipeline to connect infrastructure data, operational data, route generation, and simulation execution.

Element	Value
Simulated Area	Cour-sur-Heure, Ham-sur-Heure, Berzée, Pry
Number of Stations	4
Trip generation method	Real operational data
Simulation duration	5 days
Number of generated train routes	115
Number of complete simulated trips	115

Table 5.4: Macro-level simulation scenario summary

The result shows that all generated routes were successfully simulated. This confirms that the macro-level model is suitable for integrating real operational data when the selected network area remains compatible with the assumptions of the model. However, this result should be interpreted as a validation of station-to-station circulation, not as a complete validation of detailed railway operations. Internal station behaviour, platform allocation, switch occupation, and local conflicts are not represented at this level of abstraction.

5.3.2 Micro-Level Simulation Results

The micro-level simulation was evaluated using random trips on the reconstructed detailed railway network. The real-data micro-level scenario could not be fully executed because many station-to-station movements could not be mapped onto valid paths in the reconstructed dual graph. For this reason, the result presented here focuses on the random-trip simulation.

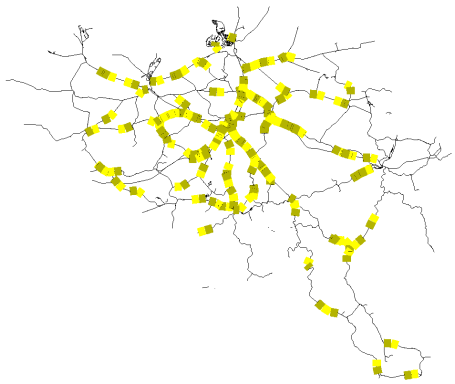

Figure 5.5: SUMO-GUI screenshot of the micro-level random trip simulation

The objective of this scenario is not to reproduce a real timetable, but to verify whether the generated micro-level infrastructure can support train circulation. Trips were generated between valid platform locations and routed through the SUMO network.

Element	Value
Simulated Area	Entire reconstructed micro-level network
Number of Stations	553
Number of Platforms	1551
Number of Tracks	26721
Trip generation method	Random trips between valid platforms
Simulation duration	1 day
Number of generated train routes	16802
Number of successfully routed trips	16802
Number of completed trips	4912

Table 5.5: Micro-level simulation scenario summary

The fact that all 16802 generated trips were successfully routed confirms that the reconstructed micro-level network is usable by SUMO for route computation. This is an important technical result, since it shows that the generated infrastructure is not only a static dataset but can also support executable train movements.

However, only 4912 trips were completed during the simulation. This difference does not necessarily indicate that the network is disconnected, because all trips were routed successfully. Instead, it reflects the limits of random trip generation on a detailed railway network. Random trips do not reproduce the full operational constraints of a real timetable, such as headways, train speeds, platform occupation, junction capacity, or temporal separation between incompatible movements. As a result, SUMO may generate warnings related to emergency braking, collisions, teleportation, or blocked insertion areas.

```
Warning: Block after insertion lane '29646_0' exceeds maximum
length(stopped searching after
edge ':2794_0' (length=20207.42m)).

Warning: Vehicle '12897' performs emergency braking on lane
'49375_0' with decel=5.00, wished=1.00, severity=1.00,
time=85488.00.

Warning: Teleporting vehicle '12897'; collision with vehicle
'10232',lane='49527_0', gap=-40.28, time=85489.00,
stage=move.
```

Overall, the micro-level simulation confirms that train movements can be executed on the reconstructed detailed infrastructure. However, it also shows that routing feasibility alone is not sufficient to obtain an operationally realistic railway simulation. Future improvements must therefore focus not only on infrastructure connectivity, but also on timetable-aware trip generation and on the improvement of the dual graph used to map real operational movements onto the detailed network.

5.4 Computational Performance

Computational performance was evaluated to estimate the cost of generating SUMO-compatible simulation scenarios from the available railway datasets. The objective is to compare the macro-level and micro-level pipelines and to identify which processing steps are the most expensive.

5.4.1 Pipeline Generation Performance

The macro-level and micro-level models differ strongly in complexity. The macro-level pipeline mainly consists of ETL processing, station-to-station network generation, trip generation from operational data, and configuration file creation. The micro-level pipeline includes additional steps such as switch detection, platform detection, track segmentation, detailed network generation, and random trip generation.

Pipeline Step	Execution Time (s)
ETL	330.92
Network generation	0.12
Trip generation	15.08
Additional file generation	0.0058
Configuration generation	0.0000
Total	331.0458

Table 5.6: Execution time of the macro-level generation pipeline for the example network

Pipeline Step	Execution Time (s)
ETL	344.09
Switch detection	1.50
Platform Detection	19.07
Track segmentation	81.90
Network generation	160.37
Trip generation	168.30
Additional file generation	0.02
Configuration generation	0.00
Total	775.25

Table 5.7: Execution time of the micro-level generation pipeline with random trip generation

The results show that the macro-level pipeline is significantly faster, with a total execution time of approximately 331 seconds, compared with approximately 775 seconds for the micro-level pipeline. This difference is expected because the macro-level model manipulates a simplified station-to-station graph, while the micro-level model processes a much larger and more detailed infrastructure.

In both cases, ETL remains an important cost because the input datasets must be loaded, cleaned, filtered, and transformed before simulation files can be generated. However, the additional cost of the micro-level pipeline mainly comes from the steps required to build a detailed simulation network. Track segmentation, network generation, and trip generation account for a large part of the additional execution time.

Overall, the performance results confirm the trade-off between abstraction and realism. The macro-level model is faster and more suitable for lightweight experiments based on real operational data. The micro-level model is computationally more expensive, but this cost is coherent with the additional infrastructure detail it provides.

5.5 Discussion

The results presented in this chapter show that the proposed pipeline can generate and simulate railway networks at two complementary levels of representation. The macro-level model provides a simplified but robust framework for integrating real operational data into SUMO, while the micro-level model provides a more detailed infrastructure representation based on tracks, platforms, switches, and segmented railway sections.

RQ1 the results confirm that heterogeneous infrastructure and operational datasets can be transformed into SUMO-compatible simulation artefacts. The generated datasets reach high coverage for stations, platforms, and tracks, showing that the preprocessing and reconstruction pipeline is able to preserve most of the available railway information. However, the results also show that coverage metrics must be interpreted carefully. In particular, the micro-level track coverage above 100% reflects the transformation of reference tracks into simulation-ready segments, rather than a direct one-to-one reproduction of the original data.

RQ2 the macro-level model validates the integration of real operational data into simulation. Train movements reconstructed from punctuality records were successfully converted into SUMO routes and simulated on a selected station chain. This confirms that the station-to-station abstraction is suitable for building a robust end-to-end workflow from real data. Its main limitation is that it cannot represent detailed infrastructure constraints such as platform occupation, switch usage, internal station routing, or local conflicts.

RQ3 the micro-level model shows that a more detailed infrastructure-based simulation is feasible, but more difficult to exploit operationally. The random-trip simulation confirms that the reconstructed network can be used by SUMO for route computation and train circulation. However, the difference between the number of routed trips and completed trips shows that routing feasibility alone is not sufficient to obtain a stable and realistic railway simulation. Random trips are useful to test connectivity, but they do not reproduce the temporal and operational constraints of a real timetable.

RQ4 the main failure point appears in the integration of real operational data at the micro level. Unlike random trips, real-data trips require each consecutive station pair in a train journey to correspond to a valid path in the reconstructed dual graph. In the current implementation, this condition was not satisfied for enough services to produce a fully exploitable real-data micro-level simulation. This issue is an important result of the thesis: it shows that a detailed railway digital twin cannot be evaluated only by the number of reconstructed elements. It must also be evaluated by its topological consistency and its ability to support operational routing.

The dual graph limitation therefore highlights a central distinction between infrastructure completeness and simulation feasibility. A network may contain almost all stations, platforms, and tracks, while still failing to reproduce real train paths if some connections are missing, incorrectly oriented, or poorly aligned with platform assignments. This explains why the macro-level model is currently more robust for real-data simulation, whereas the micro-level model provides a richer but more fragile foundation for future detailed simulations.

The computational results reinforce this trade-off. The macro-level pipeline is faster and easier to execute because it manipulates a simplified graph. The micro-level pipeline is more expensive because it includes additional processing steps such as switch detection, platform matching, track segmentation, detailed network generation, and trip generation. This additional cost is coherent with the higher level of infrastructure detail, but it also shows that scalability must be considered when moving toward larger or more realistic simulations.

Overall, the results demonstrate that the proposed framework establishes a valid foundation for a simulation-oriented railway digital twin. The macro-level model successfully validates the integration of real operational data in a simplified setting, while the micro-level model identifies the additional requirements needed for realistic infrastructure-based simulation. The main next step is therefore not only to improve coverage, but to improve route feasibility through better platform localization, stronger dual graph connectivity, improved direction handling, and timetable-aware traffic generation.

Chapter 6

Limitations

This chapter discusses the main limitations of the proposed railway digital twin simulation pipeline. Some limitations have already been introduced in Chapter 4 and Chapter 5, but they are grouped here in order to clarify the main constraints that affected the construction and execution of the macro-level and micro-level models.

The proposed framework shows that heterogeneous railway infrastructure and operational data can be transformed into SUMO-compatible simulation artefacts. However, the results also show that building a realistic railway simulation from real-world data remains difficult. The main limitations concern data availability, modelling assumptions, route reconstruction, SUMO-specific constraints, and scalability. These limitations do not invalidate the proposed methodology, but they define the boundary between the current prototype and a more complete railway digital twin.

6.1 Data Availability and Quality

A first limitation concerns the availability and precision of the data used to reconstruct the railway network. Although the available infrastructure datasets provide a strong basis for generating both macro-level and micro-level models, some information required for a fully realistic simulation is either incomplete or not directly available in a simulation-ready format. This is particularly important for the micro-level model, where the quality of the simulation depends on the correct representation of tracks, platforms, switches, and their spatial relationships.

Platform information is one example of this limitation. Platforms must be associated with specific track segments in order to create SUMO train stops. However, the available data does not always provide a direct and unambiguous mapping between each platform and the corresponding track. A proximity-based matching strategy was therefore required. While this approach provides good results in many cases, it remains an approximation and may assign platforms incorrectly in dense stations or areas with multiple parallel tracks.

Another limitation concerns operational route data. The punctuality dataset describes train passages through stations and provides temporal information such as planned and observed arrival or departure times. However, it does not provide the exact physical route followed by each train inside the railway infrastructure. In other words, the data indicates that a train moved from one station to another, but it does not specify which tracks, switches, platforms, or junctions were actually used. This absence of route-level information is one of the main obstacles to constructing a complete micro-level simulation from real data.

Because of this limitation, train paths had to be reconstructed manually or algorithmically from station sequences. This task is considerably more complex at the micro level than at the macro level. Without reference data describing the possible or historical paths used by trains, the reconstruction process must infer these paths from infrastructure topology alone. This makes the result highly dependent on the completeness and correctness of the generated graph.

Operational data also introduces limitations. The punctuality dataset provides train identifiers, station passages, planned times, observed times, and delays, but it does not provide the exact physical route followed by each train. This means that routes must be reconstructed from station sequences. At the macro level, this is generally sufficient because the network is represented as station-to-station connections. At the micro level, however, each movement must be mapped onto detailed physical paths, which requires a level of infrastructure consistency that is not always guaranteed by the available data.

6.2 Modeling Assumptions

The second limitation concerns the assumptions introduced during modelling. The macro-level model intentionally simplifies the railway network by representing stations as single nodes and railway links as direct edges. This abstraction makes the model easier to generate, faster to simulate, and more robust for integrating real operational data. However, it ignores internal station layouts, platforms, switches, parallel tracks, signalling systems, and local routing constraints.

As a result, the macro-level model cannot reproduce several important railway phenomena. It cannot model conflicts inside stations, platform occupation, switch usage, or detailed capacity constraints. It is therefore mainly appropriate for analysing station-to-station movements and validating the end-to-end simulation workflow, rather than for studying local operational behaviour.

The micro-level model reduces some of these abstractions by representing tracks, platforms, switches, and segmented infrastructure elements. However, this level of detail introduces its own assumptions. Switches are inferred from topology, platforms are detected through spatial proximity, and tracks may be segmented to improve granularity. These operations are necessary to obtain a simulation-ready network, but they also transform the original infrastructure data. The generated model should therefore be understood as an operational approximation of the railway infrastructure, not as a perfect reproduction of the real network.

6.3 Route Reconstruction and Micro-Level Simulation

The most important limitation of the current implementation concerns route reconstruction at the micro level. The macro-level model can integrate real operational data because train journeys are reconstructed as sequences of stations. In this case, a train route only needs to follow station-to-station links. This abstraction is relatively robust and allows real punctuality data to be converted into executable SUMO routes.

The micro-level model is more demanding. Each train movement between two consecutive stations must correspond to a valid path through the detailed infrastructure. This requires a continuous sequence of track segments between the selected origin and destination platforms. To achieve this, the implementation uses a dual graph representation in which track segments are treated as graph nodes and possible transitions between tracks are represented as graph edges.

In the current implementation, this approach did not produce a real-data-based micro-level simulation. Many consecutive station pairs could not be connected because no valid path was found in the generated dual graph. As a result, majority of the real train services had to be rejected before route generation. This limitation is not only an implementation issue, but also an important methodological finding.

The failure of the dual graph highlights the difference between infrastructure coverage and simulation feasibility. A generated network may contain most stations, platforms, tracks, and switches, but still fail to reproduce real train movements if some connections are missing, incorrectly oriented, or not aligned with the detected platforms. In other words, high dataset coverage does not guarantee that the network is operationally routable.

This limitation has direct implications for railway digital twin construction. A detailed digital twin cannot be evaluated only by counting reconstructed infrastructure elements. It must also be evaluated by its ability to support valid train paths under realistic operational constraints. Future work should therefore focus on improving dual graph connectivity, direction handling, platform assignment, and station-to-station path reconstruction.

6.4 Simulation Constraints in SUMO

Another limitation comes from the use of SUMO for railway simulation. SUMO provides a flexible and scriptable simulation environment, which makes it well suited for automated network generation and experimental workflows. However, it is not originally a railway-only simulator. Several railway-specific mechanisms are therefore not automatically represented and must be simplified or explicitly added.

In particular, detailed signalling, interlocking, dispatching, block occupation, platform assignment, and conflict resolution are not fully modelled in the current implementation. This affects the realism of the simulation, especially at the micro level. Random trips can be routed and executed in SUMO, but they do not necessarily respect the temporal structure, operational priorities, or safety constraints of real railway traffic.

SUMO-related warnings or execution issues must therefore be interpreted carefully. Some problems may come from the generated infrastructure, while others may result from unrealistic traffic demand, missing railway-specific control logic, or simplified operational assumptions. The current implementation uses SUMO primarily as a flexible simulation platform for testing network generation, route feasibility, and data integration, rather than as a complete railway traffic management system.

6.5 Scalability and Computational Complexity

The final limitation concerns scalability and computational complexity. The macro-level model is relatively lightweight because it manipulates a reduced number of entities. Stations are represented as nodes, railway links as edges, and real train movements can be reconstructed at station level. This makes the macro-level pipeline suitable for rapid experimentation and for validating the integration of operational data.

The micro-level model is more computationally expensive. It requires processing a larger number of infrastructure elements and performing additional steps such as switch detection, platform matching, track segmentation, dual graph construction, trip generation, and routing. This additional cost is expected because the model provides a more detailed representation of the infrastructure, but it also limits the ease with which large-scale scenarios can be generated and tested.

Scalability is therefore linked to the trade-off between abstraction and realism. The macro-level model is more robust and easier to execute, but less detailed. The micro-level model is more realistic, but more sensitive to data quality, routing inconsistencies, and computational cost. A future version of the framework could combine both approaches by using the macro-level model for large-scale operational flows and the micro-level model only in selected areas where detailed infrastructure behaviour is required.

Overall, the limitations identified in this chapter show that the proposed framework should be interpreted as a foundation for a railway digital twin rather than as a complete operational system. The macro-level model validates the integration of real data into simulation, while the micro-level model reveals the additional challenges introduced by detailed infrastructure reconstruction. In particular, the dual graph issue shows that the next step is not only to generate more infrastructure elements, but to ensure that the generated network is topologically consistent and operationally usable.

Chapter 7

Conclusion

This thesis investigated the construction of a simulation-oriented digital twin for railway network simulation using SUMO and heterogeneous railway datasets. The objective was not to build a complete real-time digital twin, but to study the foundations required to transform infrastructure and operational data into executable simulation models.

The work was structured around two complementary levels of representation. The macro-level model represents the railway network as a station-to-station graph and focuses on the integration of real operational data. The micro-level model represents the infrastructure in more detail through tracks, platforms, switches, and routing structures. Together, these two approaches make it possible to evaluate the trade-off between simplicity, scalability, infrastructure realism, and simulation feasibility.

7.1 Summary of Contributions

The first contribution of this thesis is the design and implementation of a data processing pipeline able to transform railway datasets into SUMO-compatible artefacts. This includes the extraction and preprocessing of infrastructure and operational data, the generation of simulation networks, the construction of train routes, and the conversion of simulation outputs into analysable datasets.

The second contribution is the development of a macro-level railway simulation model. This model abstracts the railway network as stations connected by railway links. Although simplified, it enables the integration of real punctuality data and validates the complete workflow from raw data to simulation execution.

The third contribution is the development of a micro-level railway infrastructure model. This model introduces more detailed infrastructure processing steps, including switch detection, platform matching, track segmentation, and dual graph construction. It provides a richer representation of the railway network and makes it possible to test train circulation on a more detailed infrastructure.

Finally, this work contributes by identifying the main technical challenges involved in building railway digital twins from real-world data. These challenges include data alignment, infrastructure completeness, platform assignment, route reconstruction, SUMO-specific constraints, and computational scalability.

7.2 Key Findings

Regarding **RQ1**, the results show that heterogeneous railway infrastructure and operational datasets can be transformed into SUMO-compatible simulation components through a structured pipeline. The generated datasets provide a strong basis for simulation, especially at the station and platform level.

Regarding **RQ2**, the macro-level model proved to be the most robust representation for integrating real operational data. Since train movements can be reconstructed as station-to-station trajectories, the macro model supports real-data simulation on selected chains of stations. It therefore provides an effective first validation of the digital twin workflow.

Regarding **RQ3**, the micro-level model shows that detailed infrastructure reconstruction is feasible and useful. The generated network can represent tracks, switches, platforms, and segmented railway sections. Random-trip simulations confirm that the infrastructure can support train circulation in SUMO.

Regarding **RQ4**, the main limitation appears when real operational data is integrated into the micro-level model. Real train services require continuous and valid paths between consecutive stations. In the current implementation, this condition is not always satisfied because the reconstructed dual graph does not always provide valid paths between station pairs. This result is important because it shows that high dataset coverage does not necessarily imply operational validity. A railway digital twin must therefore be evaluated not only by the number of reconstructed elements, but also by its ability to support realistic train routes.

Overall, the main finding is that macro-level and micro-level models are complementary. The macro-level model is more robust and scalable for real-data simulation, while the micro-level model provides a richer basis for detailed infrastructure analysis but requires stronger topological consistency.

7.3 Practical Implications

The proposed framework shows that existing railway data can be transformed into executable simulation scenarios without manually building the entire network. This is relevant for researchers and railway stakeholders interested in simulation-based analysis, data integration, or early-stage digital twin development.

The macro-level model can be used for preliminary traffic analysis, selected line simulation, and validation of operational data integration. Its simplified structure makes it easier to execute and less sensitive to missing detailed infrastructure information.

The micro-level model is more limited in its current form, but it provides a foundation for future analyses involving platforms, switches, local bottlenecks, routing conflicts, and infrastructure capacity. Its main value lies in showing where the transition from a structural infrastructure model to an operational simulation model becomes difficult.

The work also confirms that SUMO can be used as a flexible platform for railway digital twin prototyping. However, several railway-specific elements, such as signalling, interlocking, dispatching, platform allocation, and conflict management, require additional modelling beyond the standard SUMO workflow.

7.4 Future directions

The work presented in this thesis provides a first implementation of a railway digital twin pipeline based on real infrastructure and operational data. Future work should focus both on improving the existing implementation and on extending the framework toward more advanced decision-support functionalities.

7.4.1 Improving Pathfinding and Micro-Level Routing

The first priority is to improve the micro-level pathfinding process. The current implementation relies on a dual graph representation, where track segments are treated as graph nodes and transitions between tracks define the possible movements. This approach is useful, but the results show that it is not sufficient in its current form to reconstruct real train services reliably.

Future work should therefore investigate alternative or complementary routing methods. One possibility would be to construct a directed infrastructure graph closer to SUMO's internal edge representation and perform pathfinding directly on this graph. Another possibility would be to combine the dual graph with additional validation rules based on track direction, switch connectivity, platform accessibility, and known station-to-station connections. A hybrid routing strategy could also be considered, where macro-level paths are first used to constrain the search space before computing detailed micro-level routes.

Improving this part of the code would also require better diagnostics. Instead of only detecting that no path exists, the pipeline should identify why the path failed: missing edge, wrong direction, disconnected platform, switch inconsistency, or incomplete station mapping. This would make debugging easier and would help transform routing failures into actionable information.

7.4.2 Improving Existing Code Structure

Another important direction concerns the maintainability of the implementation. The current code validates the main concepts of the thesis, but future versions should make the pipeline more modular. Data extraction, preprocessing, network generation, platform matching, routing, trip generation, simulation execution, and output analysis could be separated into independent modules with clearer interfaces.

This would make the framework easier to test, extend, and reuse. It would also allow each component to be evaluated independently. For example, platform matching could be tested without running a full simulation, and routing algorithms could be compared on the same infrastructure graph. Adding automated validation tests would also help detect regressions when modifying the pipeline.

7.4.3 Counterfactual Simulation and What-If Scenarios

A further direction concerns counterfactual simulation and what-if scenarios. The current framework mainly reproduces existing infrastructure and train movements. A future version could allow users to modify parts of the network and observe the impact of these changes on railway traffic.

For example, it could be possible to simulate the closure of a station, the removal of a track section, a modification of train frequency, or a change in departure times. At the macro level, this could be implemented by modifying station-to-station connections. At the micro level, specific platforms, switches, or track segments could be made unavailable. This would make the digital twin useful as an exploratory tool for testing operational decisions before applying them to the real network.

7.4.4 Tracks Works Planning

Track works planning is another promising extension. Maintenance operations often require temporary closures, reduced capacity, speed restrictions, or rerouting. A simulation-based digital twin could help evaluate the impact of these interventions before they take place.

In the current framework, this could be implemented by adding availability constraints to infrastructure elements. At the macro level, railway connections could be removed or penalized. At the micro level, specific tracks, platforms, or switches could be marked as unavailable during a given time window. This would help compare maintenance strategies, identify bottlenecks, and reduce the operational impact of planned works.

7.4.5 Real-Time Forecasting and Incident Propagation

Another future direction is the integration of real-time forecasting and incident propagation. The current implementation is mainly based on historical and static data. A future version could integrate live or near-real-time information, such as train delays or incidents, and use it to estimate the future state of the network.

This extension could combine the simulation pipeline with statistical or machine learning models trained on punctuality data. The macro-level model could be used to study delay propagation between stations, while the micro-level model could help analyse the influence of local infrastructure constraints. Such an extension would move the framework closer to an operational digital twin for monitoring and proactive traffic management.

7.4.6 Framework Generalization

A final direction concerns the generalization of the framework. The current implementation is adapted to the Belgian railway datasets used in this thesis. A more general version should separate the core modeling logic from dataset-specific preprocessing steps.

This could be achieved by introducing modular data adapters that convert different railway datasets into a common internal representation. Depending on the available data, the framework could then generate a macro-level model, a micro-level model, or a hybrid representation. This would make the approach easier to transfer to other railway networks and would improve its long-term usability.

7.5 Final Remarks

This thesis shows that building a railway digital twin is not only a simulation problem, but also a data integration and infrastructure modelling problem. The results demonstrate that a complete pipeline can be constructed from real railway datasets and executed in SUMO, especially at the macro level. They also show that moving toward a detailed micro-level representation introduces new challenges, particularly in route reconstruction and topological consistency.

The most important lesson of this work is that infrastructure coverage alone is not sufficient. A generated railway network must also be operationally usable: platforms must be reachable, track directions must be coherent, switches must preserve connectivity, and real train paths must be routable. The failure of the real-data micro-level simulation is therefore not only a limitation, but also a valuable finding. It identifies the next technical obstacle to solve in order to move from a structural digital representation toward a more complete operational railway digital twin.

Overall, this work provides a first foundation for simulation-oriented railway digital twin development. It validates the feasibility of the macro-level approach, demonstrates the potential of the micro-level approach, and identifies the improvements required to make future versions more realistic, robust, and useful for railway analysis.

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